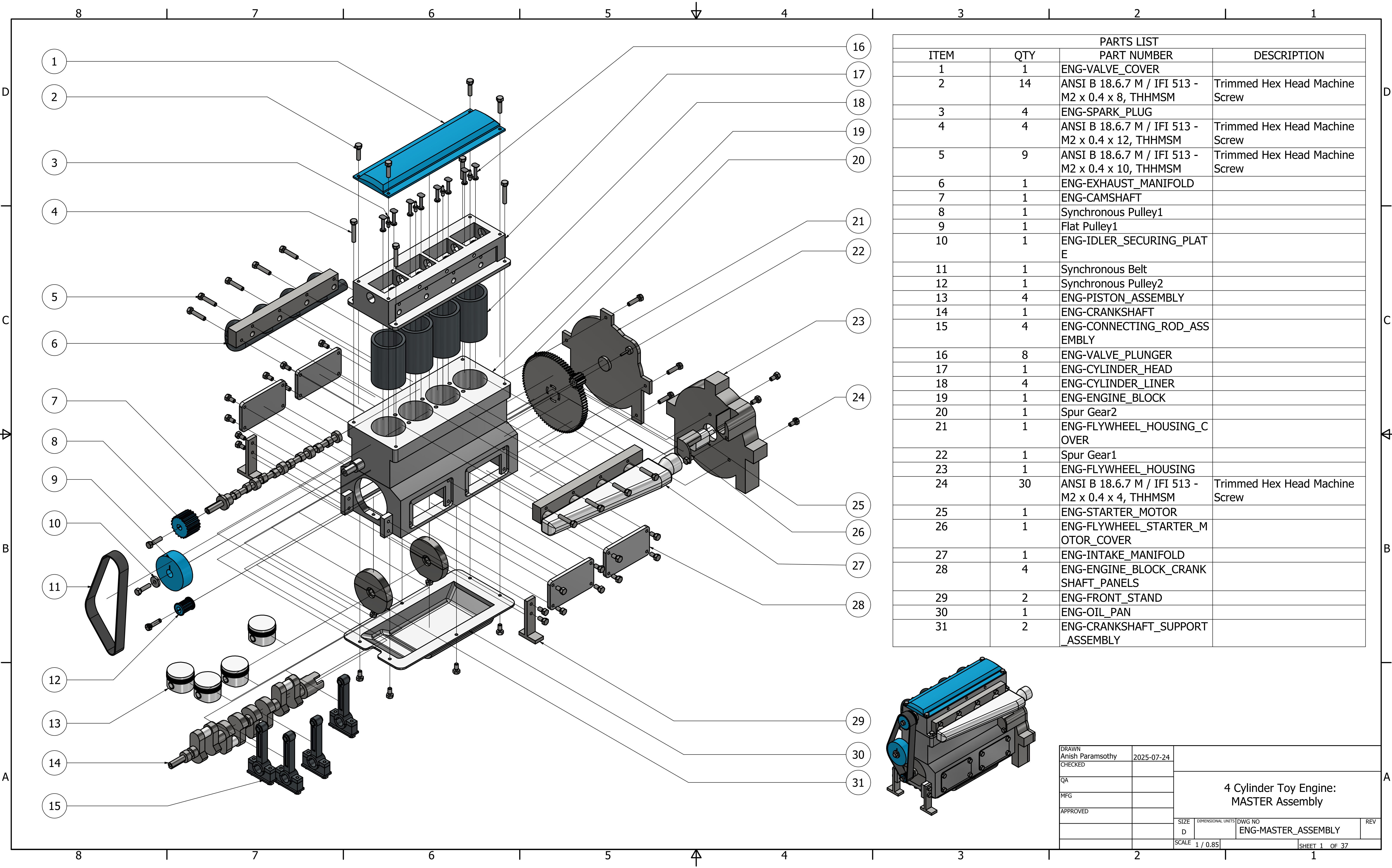
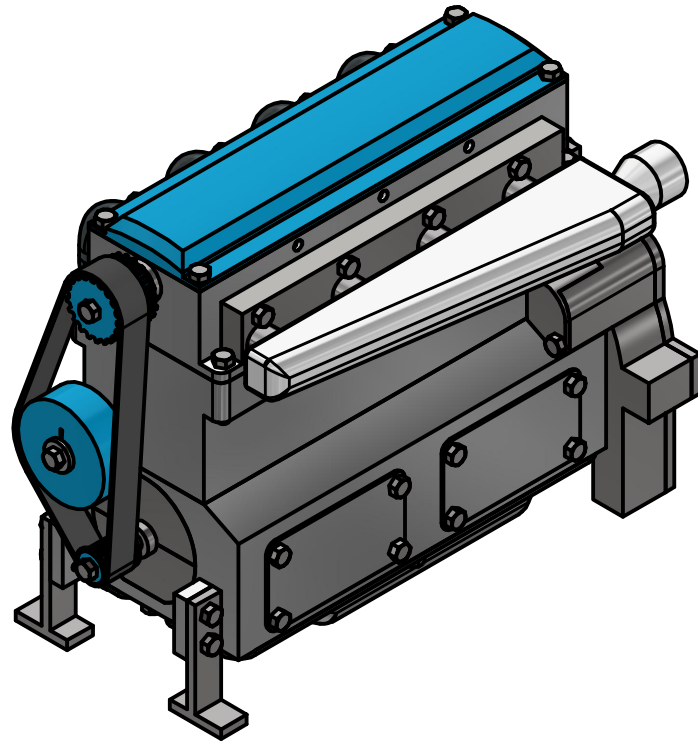
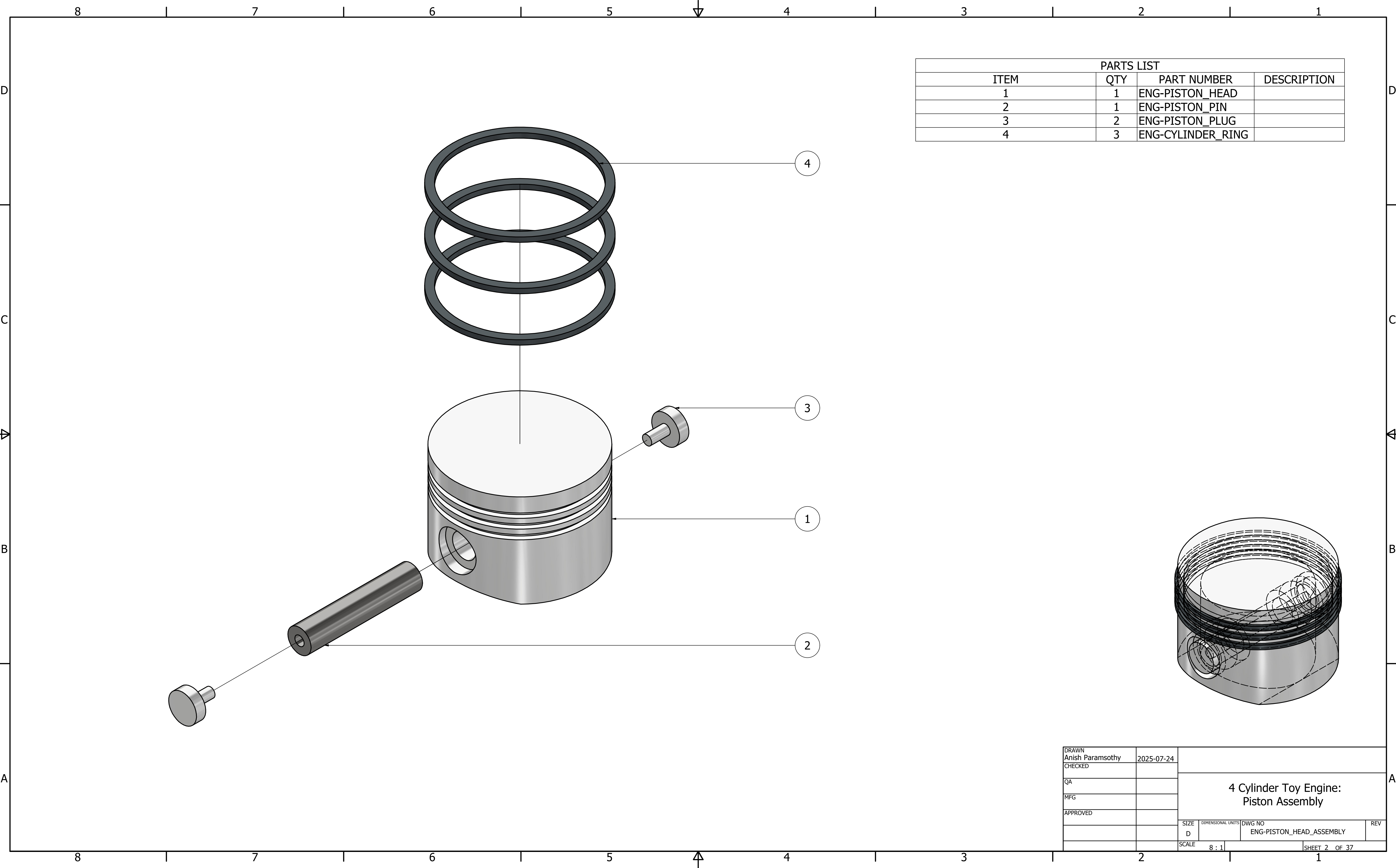




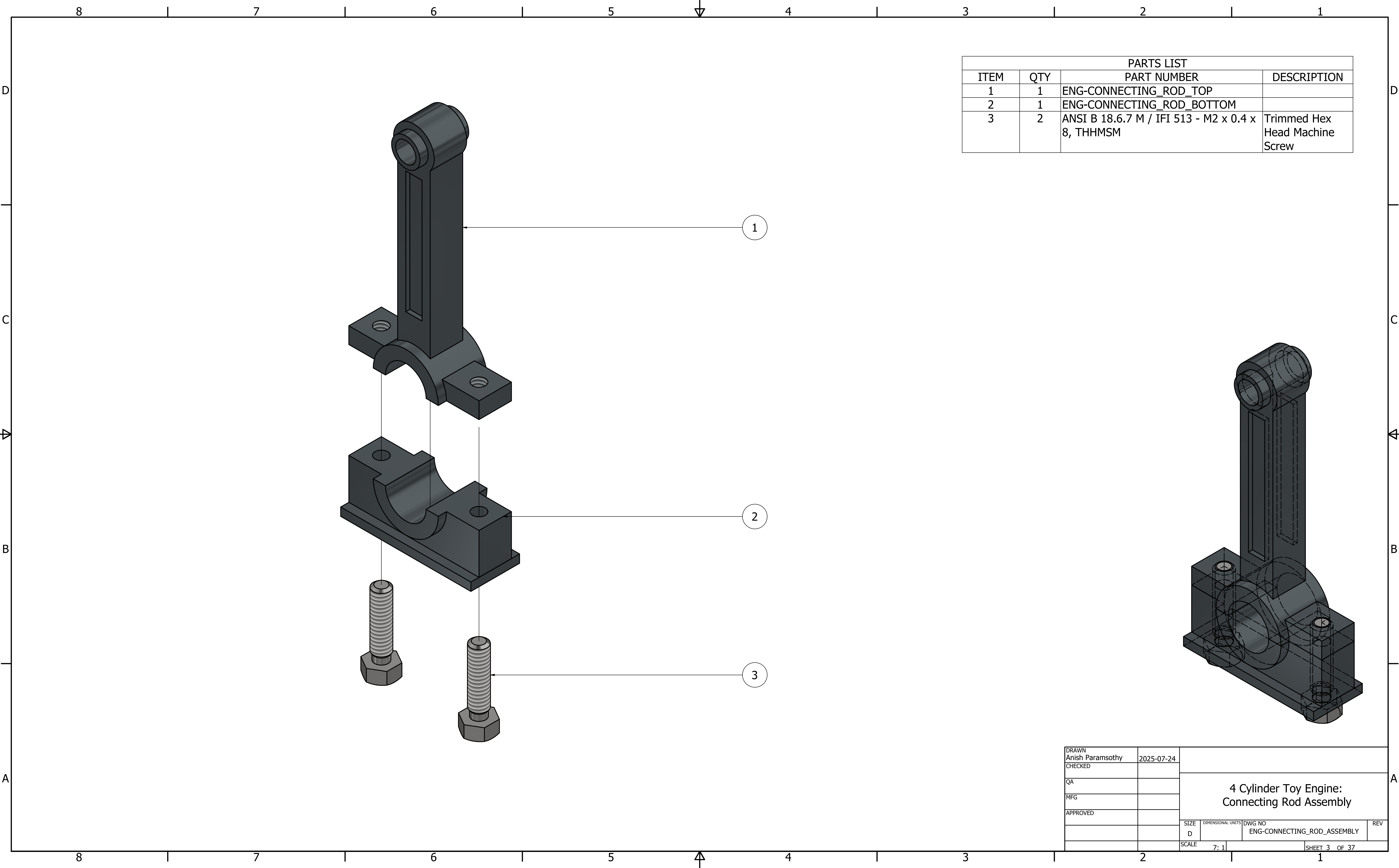
PARTS LIST			
ITEM	QTY	PART NUMBER	DESCRIPTION
1	1	ENG-VALVE_COVER	
2	14	ANSI B 18.6.7 M / IFI 513 - M2 x 0.4 x 8, THHMSM	Trimmed Hex Head Machine Screw
3	4	ENG-SPARK_PLUG	
4	4	ANSI B 18.6.7 M / IFI 513 - M2 x 0.4 x 12, THHMSM	Trimmed Hex Head Machine Screw
5	9	ANSI B 18.6.7 M / IFI 513 - M2 x 0.4 x 10, THHMSM	Trimmed Hex Head Machine Screw
6	1	ENG-EXHAUST_MANIFOLD	
7	1	ENG-CAMSHAFT	
8	1	Synchronous Pulley1	
9	1	Flat Pulley1	
10	1	ENG-IDLER_SECURING_PLATE	
11	1	Synchronous Belt	
12	1	Synchronous Pulley2	
13	4	ENG-PISTON_ASSEMBLY	
14	1	ENG-CRANKSHAFT	
15	4	ENG-CONNECTING_ROD_ASSEMBLY	
16	8	ENG-VALVE_PLUNGER	
17	1	ENG-CYLINDER_HEAD	
18	4	ENG-CYLINDER_LINER	
19	1	ENG-ENGINE_BLOCK	
20	1	Spur Gear2	
21	1	ENG-FLYWHEEL_HOUSING_COVER	
22	1	Spur Gear1	
23	1	ENG-FLYWHEEL_HOUSING	
24	30	ANSI B 18.6.7 M / IFI 513 - M2 x 0.4 x 4, THHMSM	Trimmed Hex Head Machine Screw
25	1	ENG-STARTER_MOTOR	
26	1	ENG-FLYWHEEL_STARTER_MOTOR_COVER	
27	1	ENG-INTAKE_MANIFOLD	
28	4	ENG-ENGINE_BLOCK_CRANKSHAFT_PANELS	
29	2	ENG-FRONT_STAND	
30	1	ENG-OIL_PAN	
31	2	ENG-CRANKSHAFT_SUPPORT_ASSEMBLY	

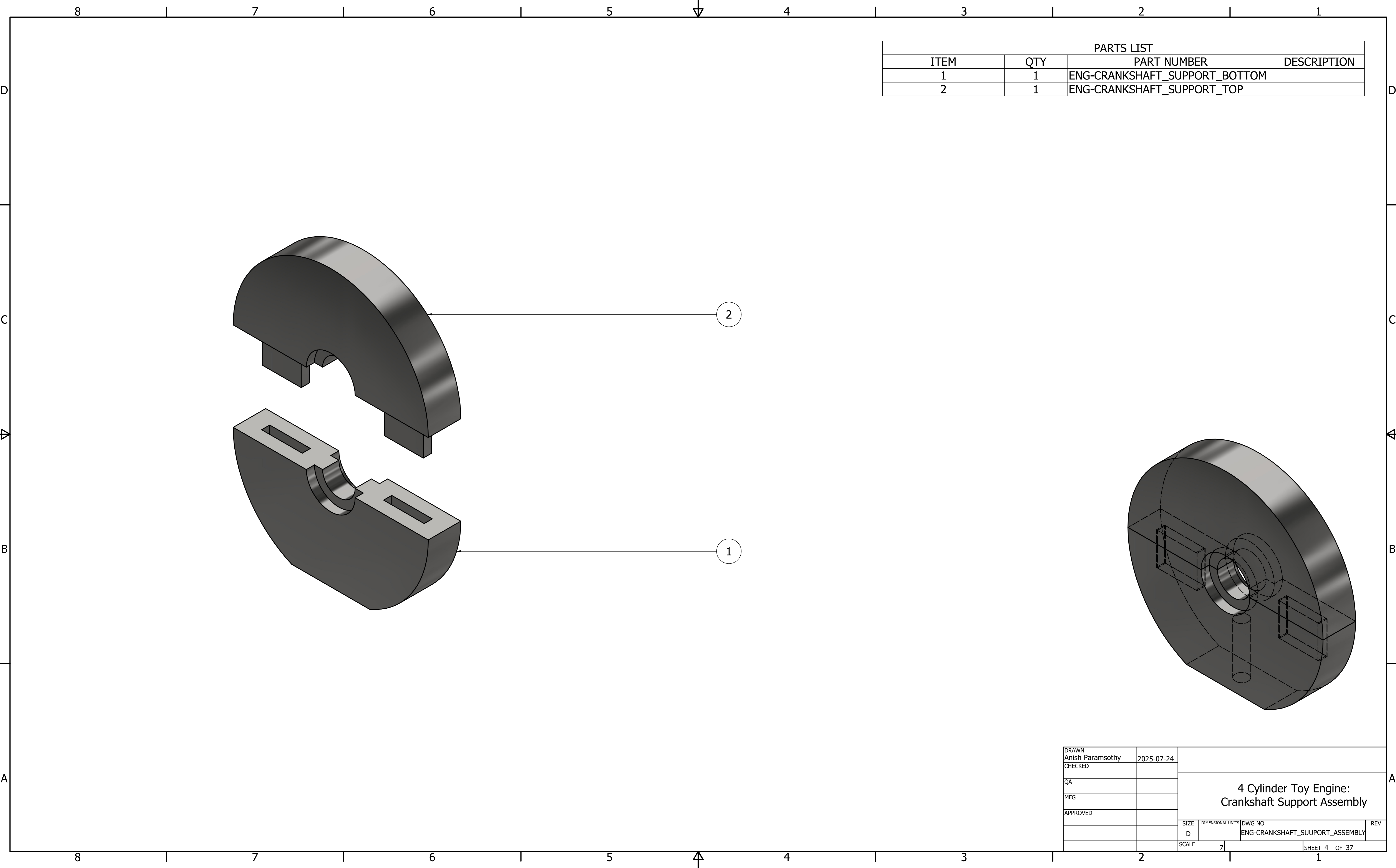


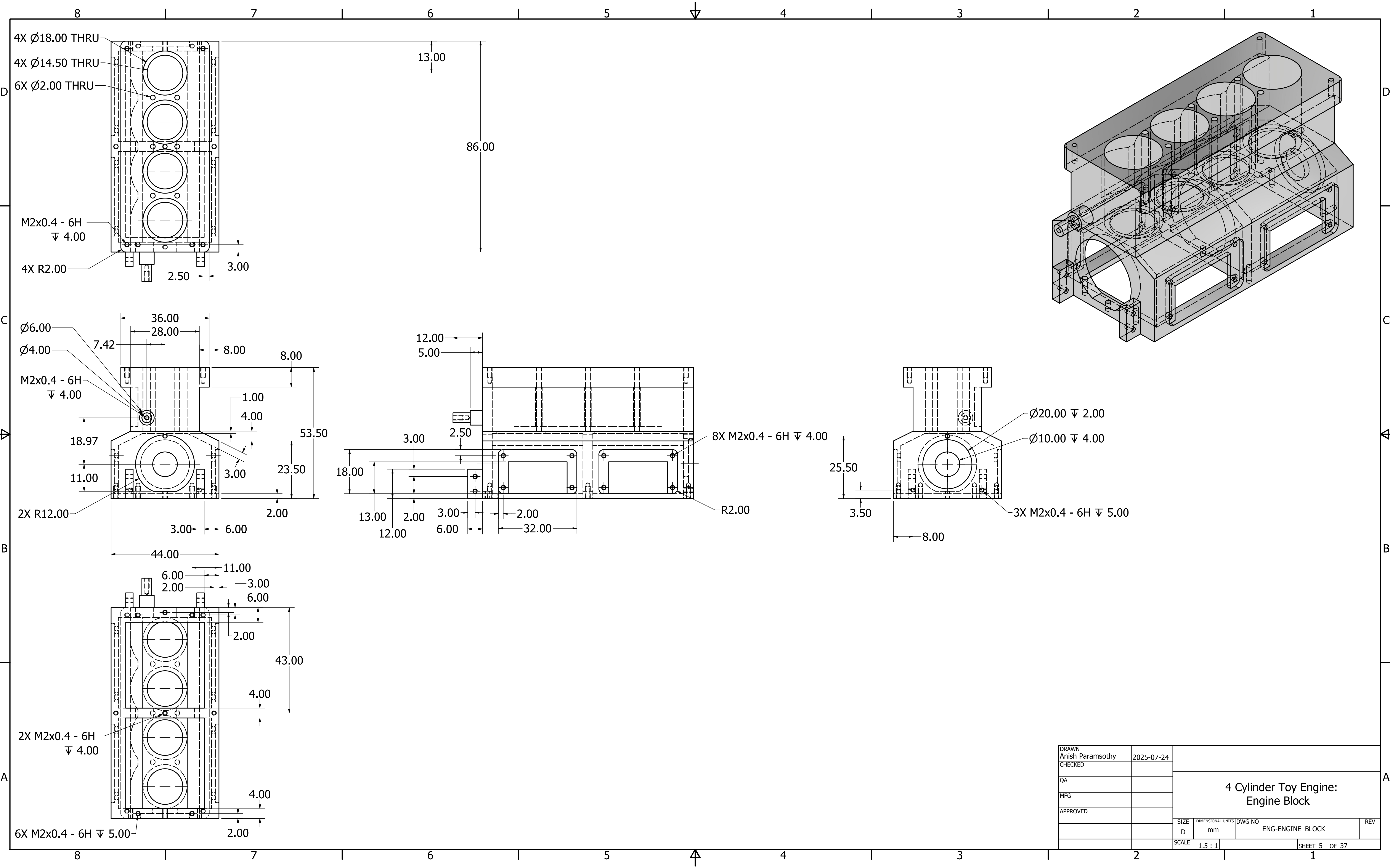
DRAWN Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine: MASTER Assembly		
CHECKED				
QA				
MFG				
APPROVED		SIZE D	DIMENSIONAL UNITS DWG NO ENG-MASTER_ASSEMBLY	REV
		SCALE 1 / 0.85	SHEET 1 OF 37	



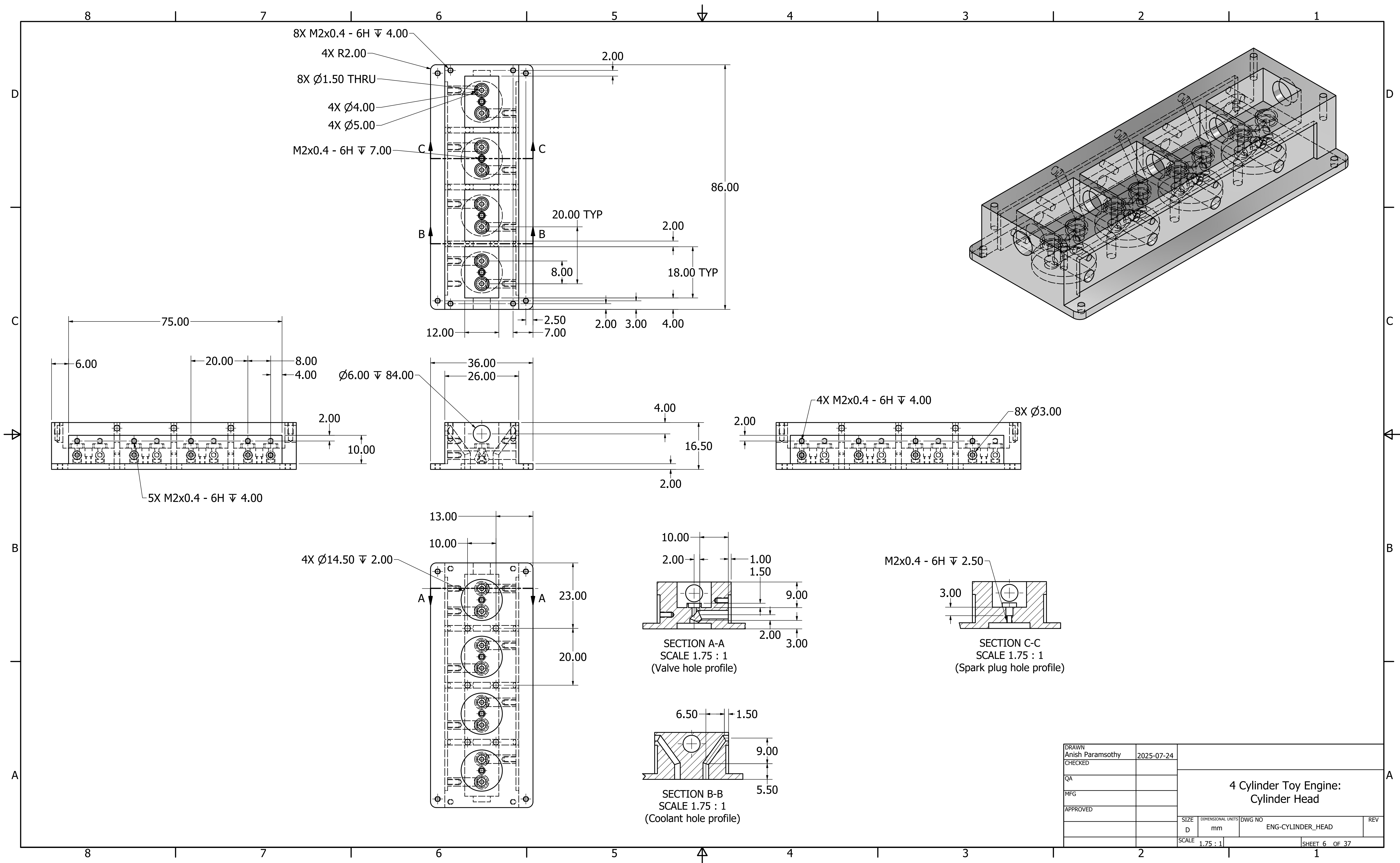
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine: Piston Assembly			
CHECKED						
QA						
MFG						
APPROVED						
			SIZE	DIMENSIONAL UNITS	DWG NO	REV
			D		ENG-PISTON_HEAD_ASSEMBLY	
			SCALE	8 : 1		SHEET 2 OF 37

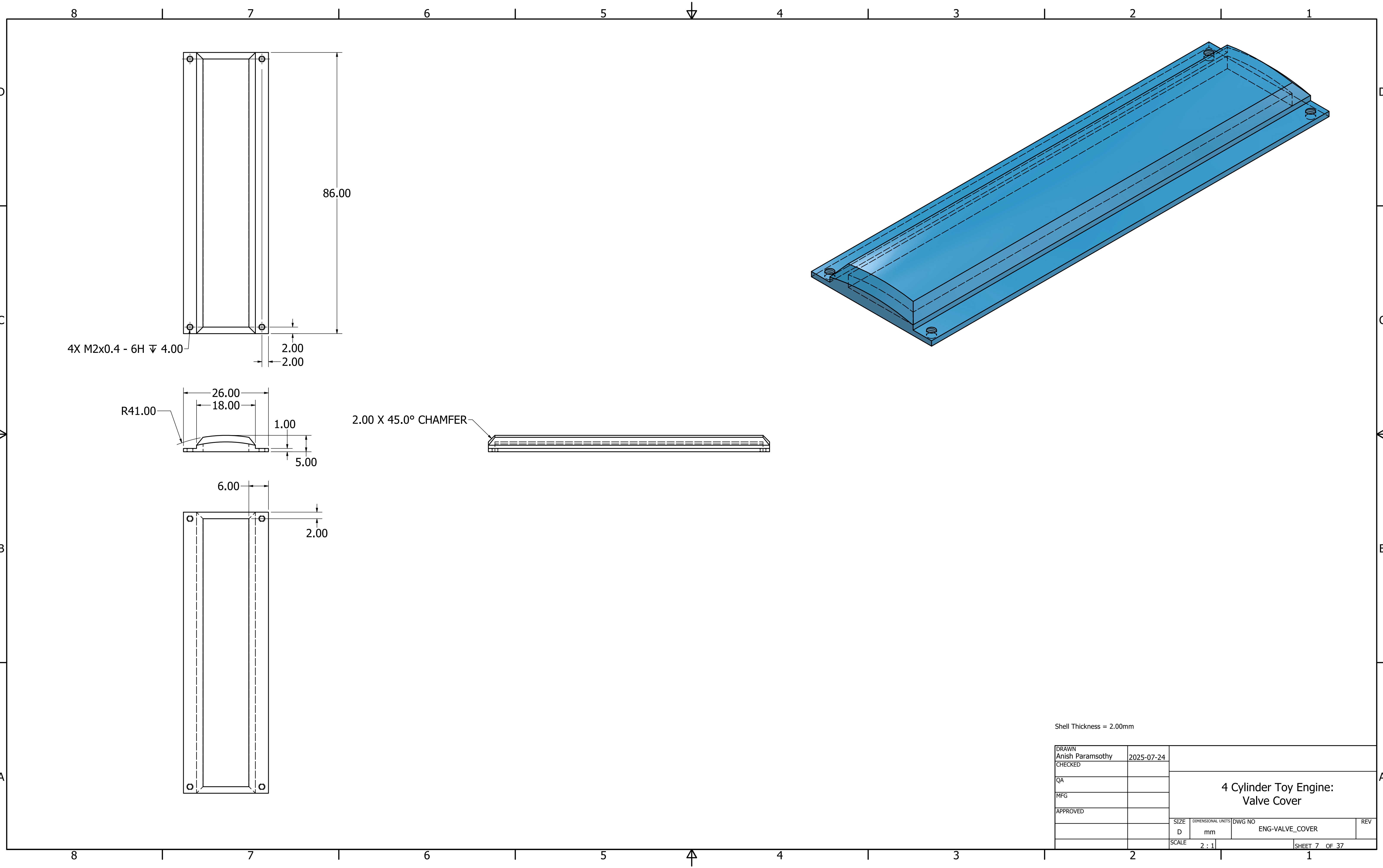






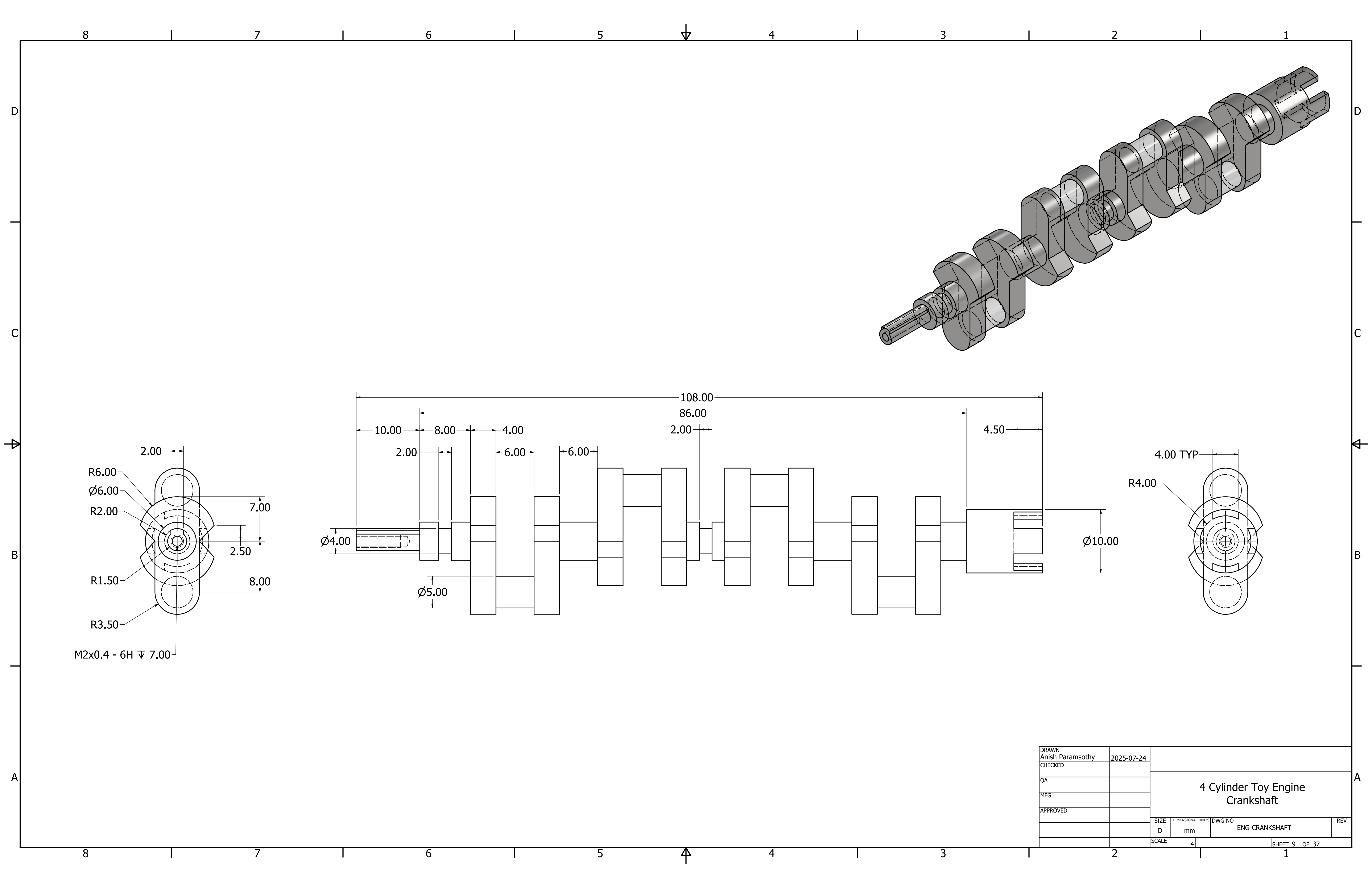
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine: Engine Block		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-ENGINE_BLOCK
			SCALE	1.5 : 1	SHEET 5 OF 37

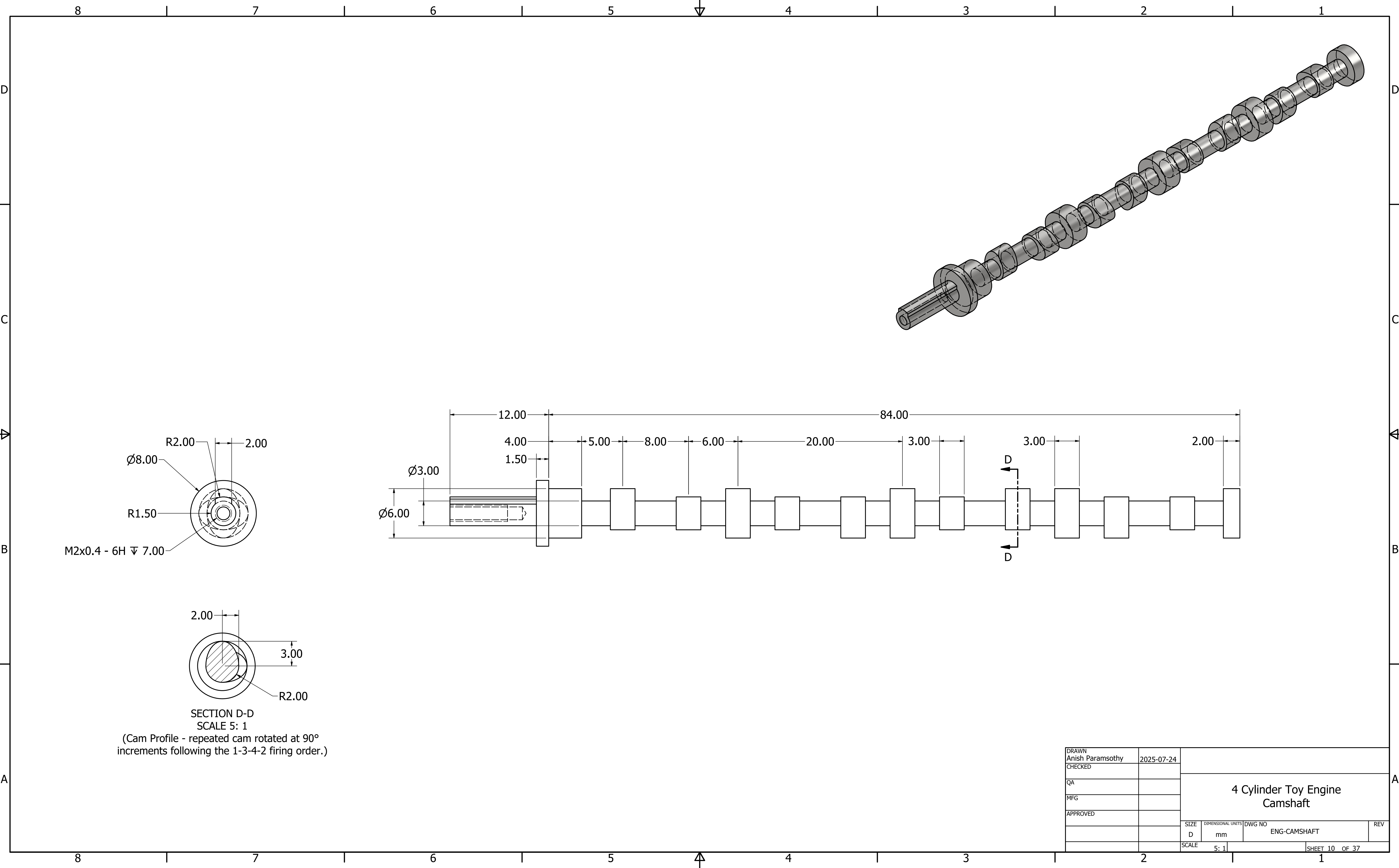


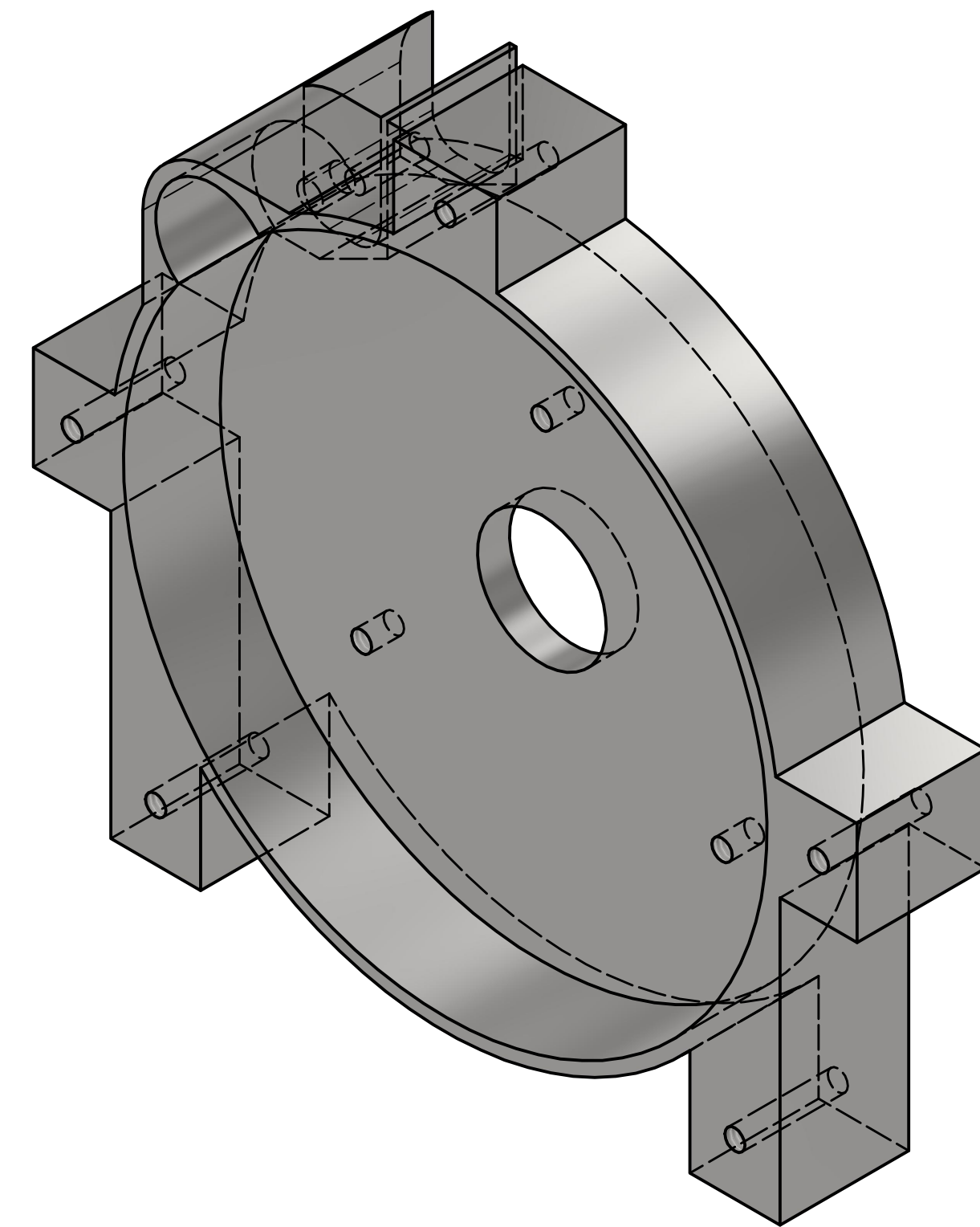
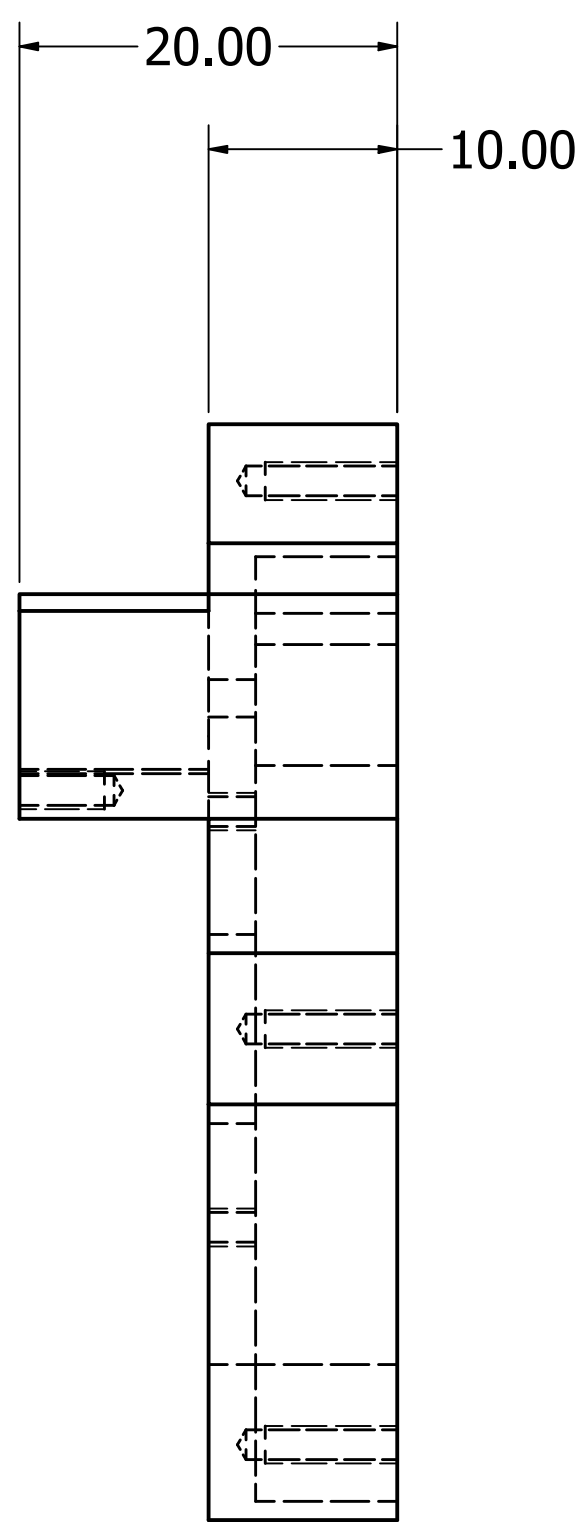


Shell Thickness = 2.00mm

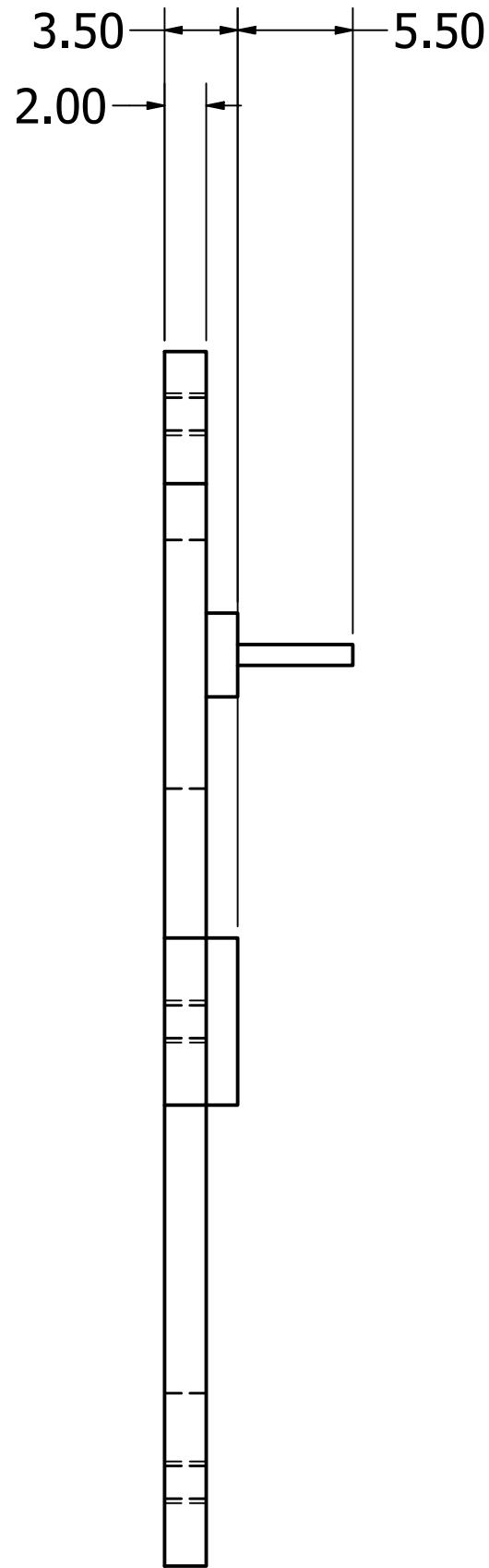
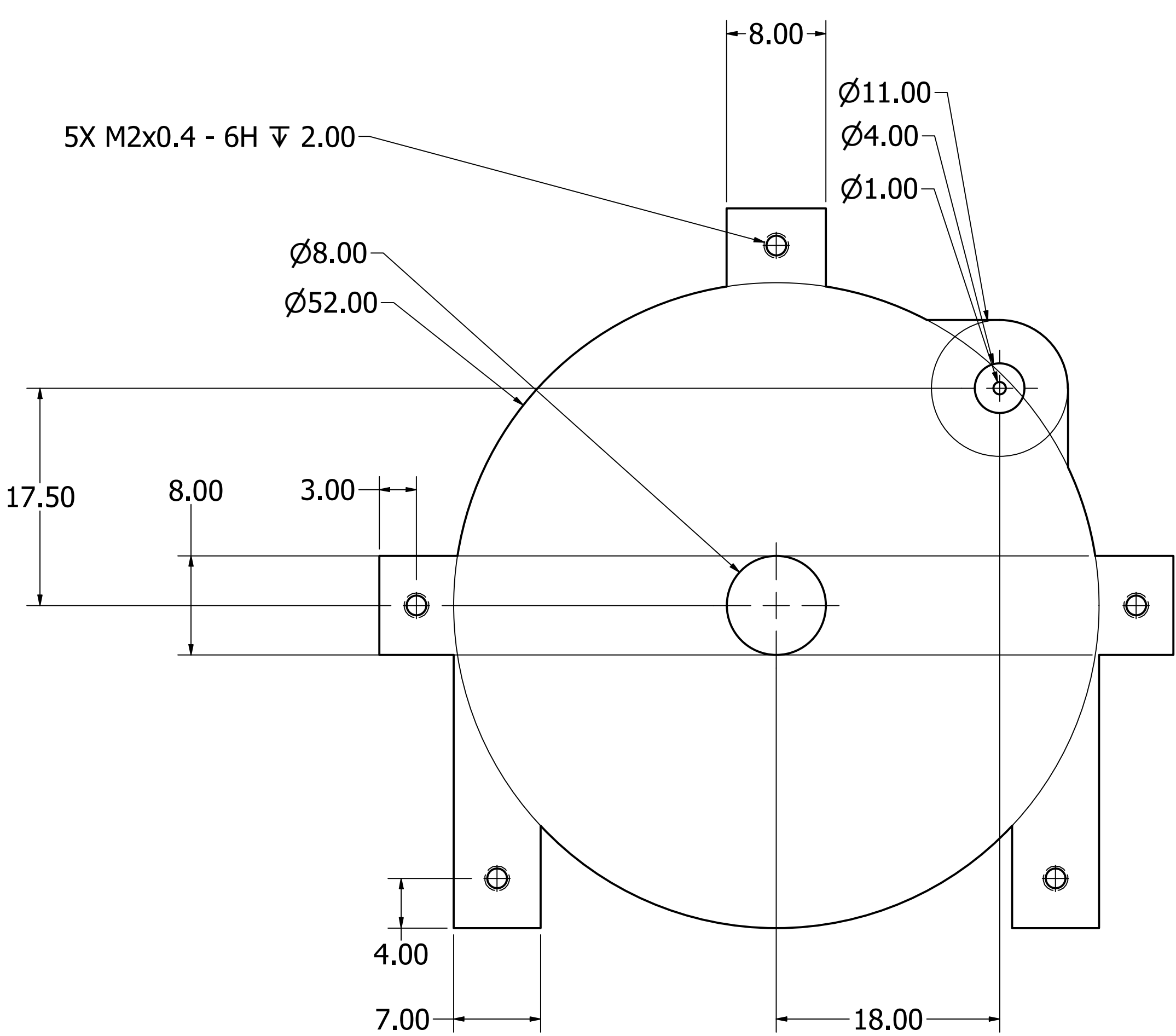
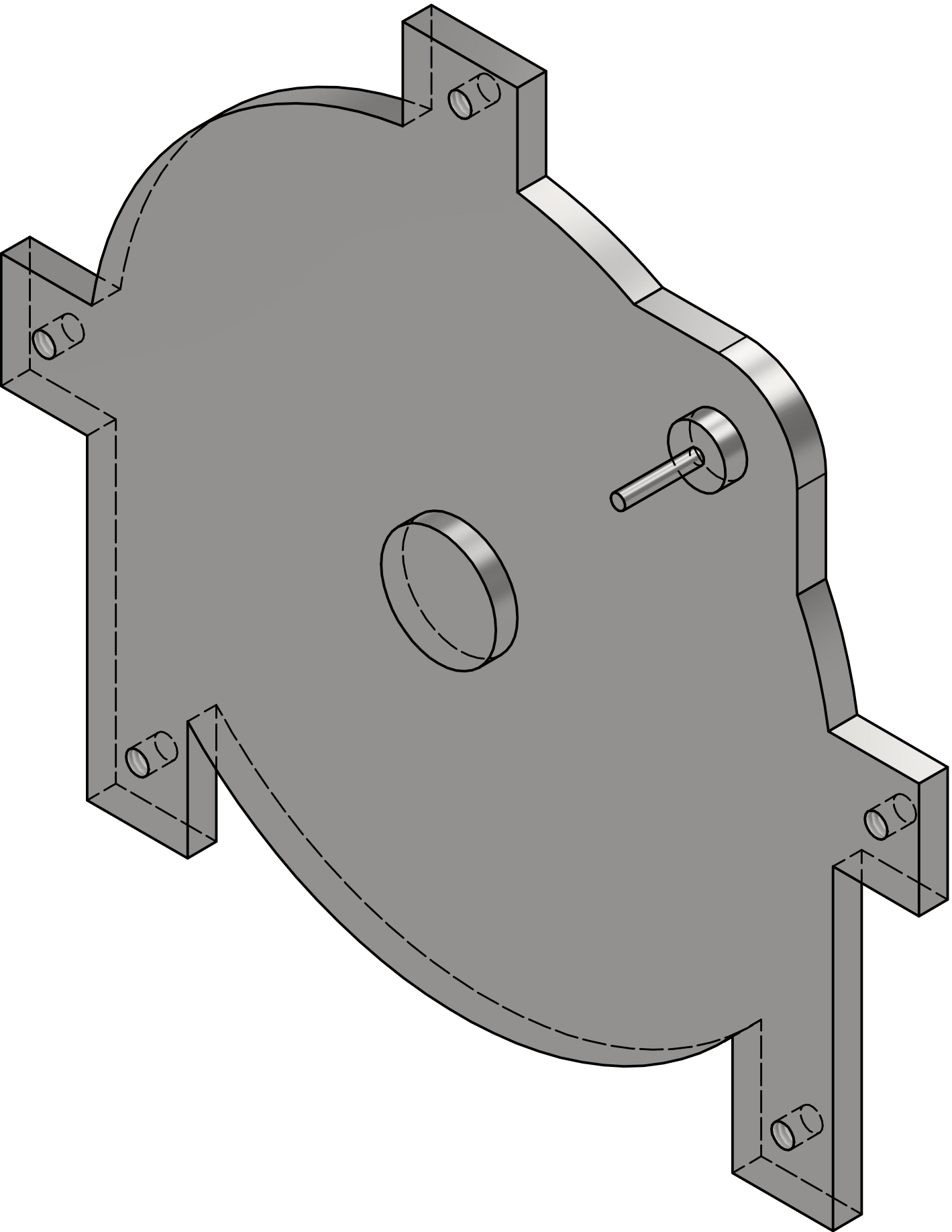
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine: Valve Cover		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-VALVE_COVER
			SCALE	2 : 1	SHEET 7 OF 37



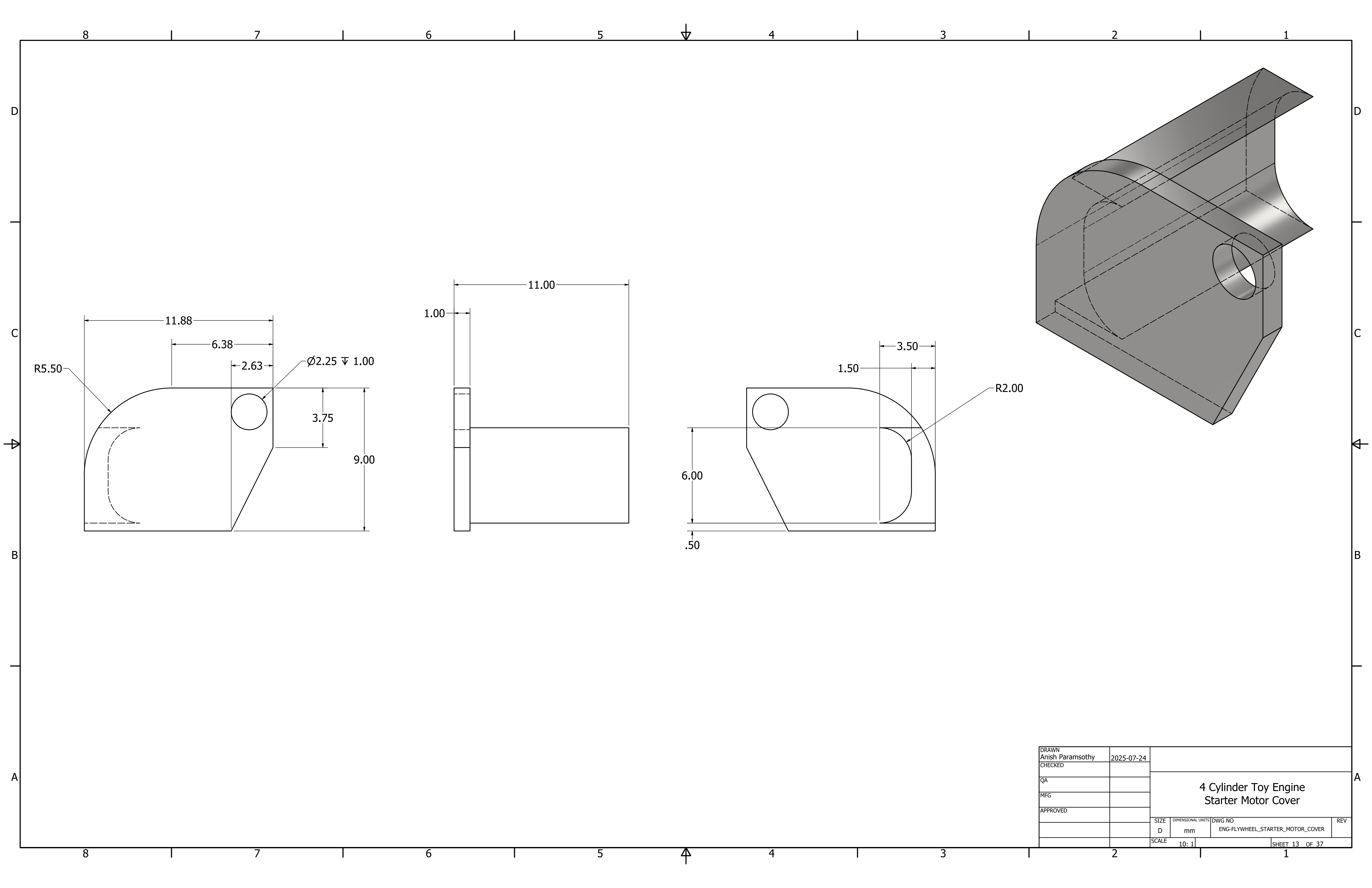




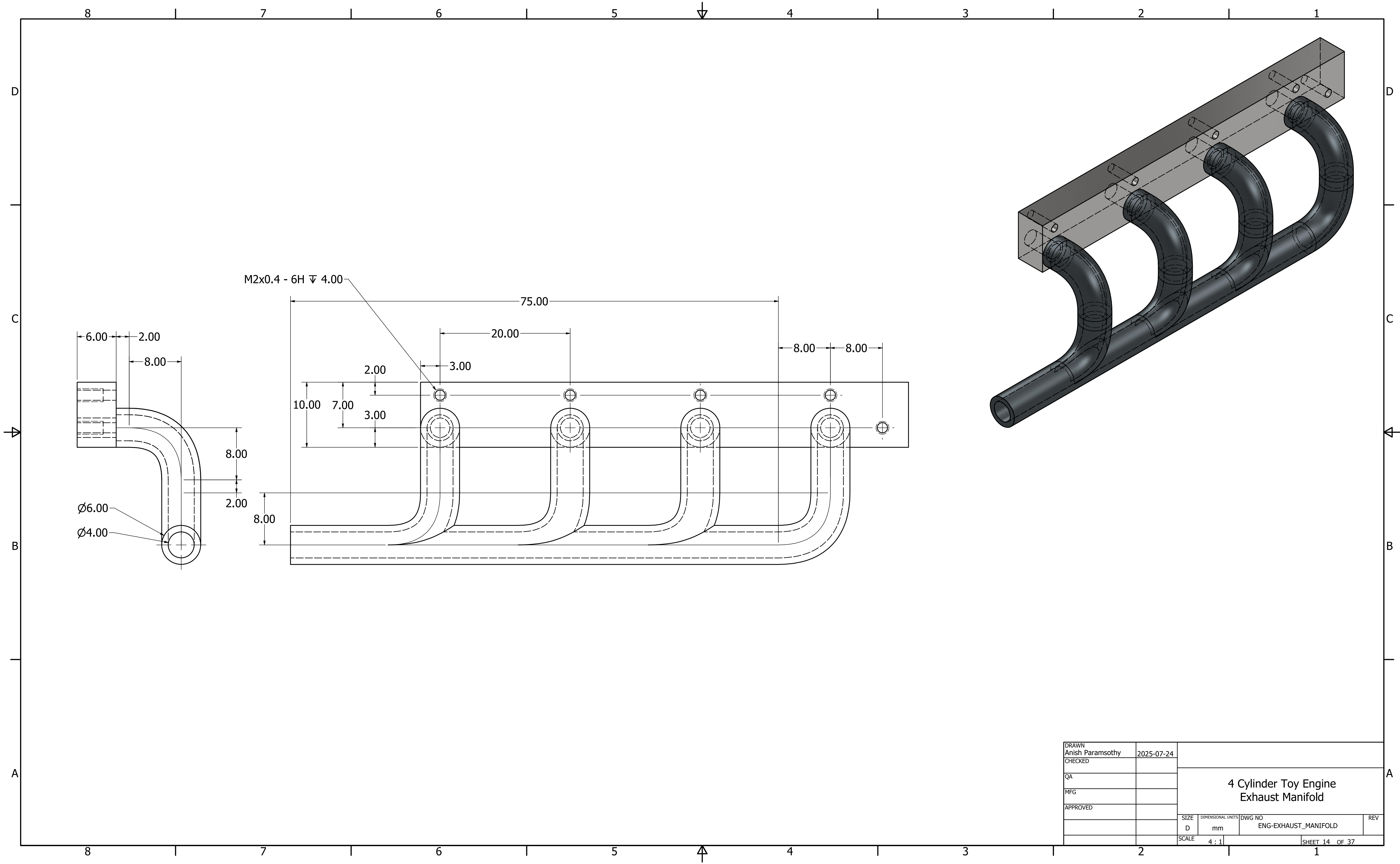
DRAWN Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Flywheel Housing				
CHECKED						
QA						
MFG						
APPROVED						
		SIZE D	DIMENSIONAL UNITS mm	DWG NO	ENG-FLYWHEEL_HOUSING	RE
		SCALE	2.5: 1		SHEET 11	OF 37

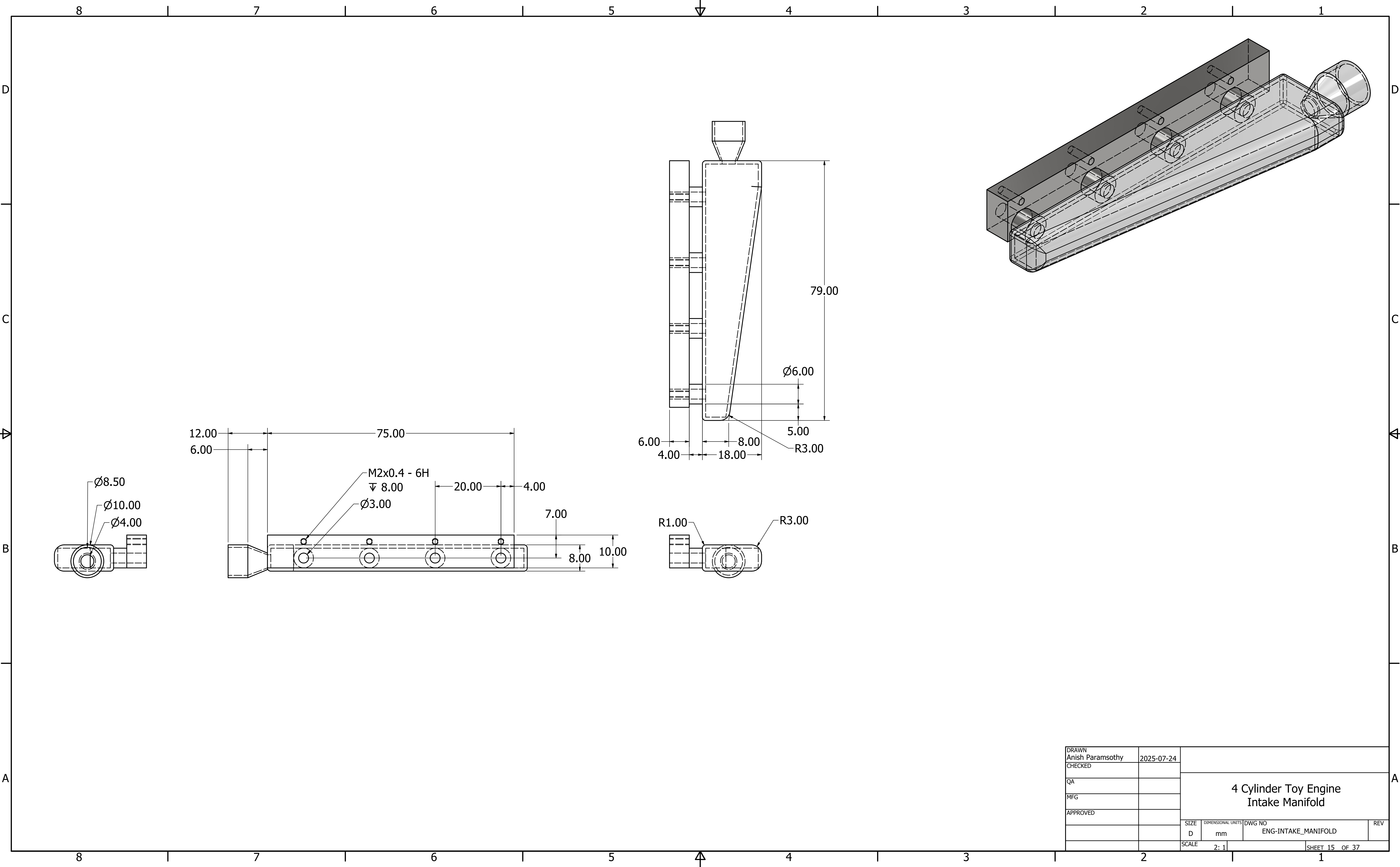


DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Flywheel Housing Cover		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-FLYWHEEL_HOUSING_COVER
			SCALE	3: 1	SHEET 12 OF 37

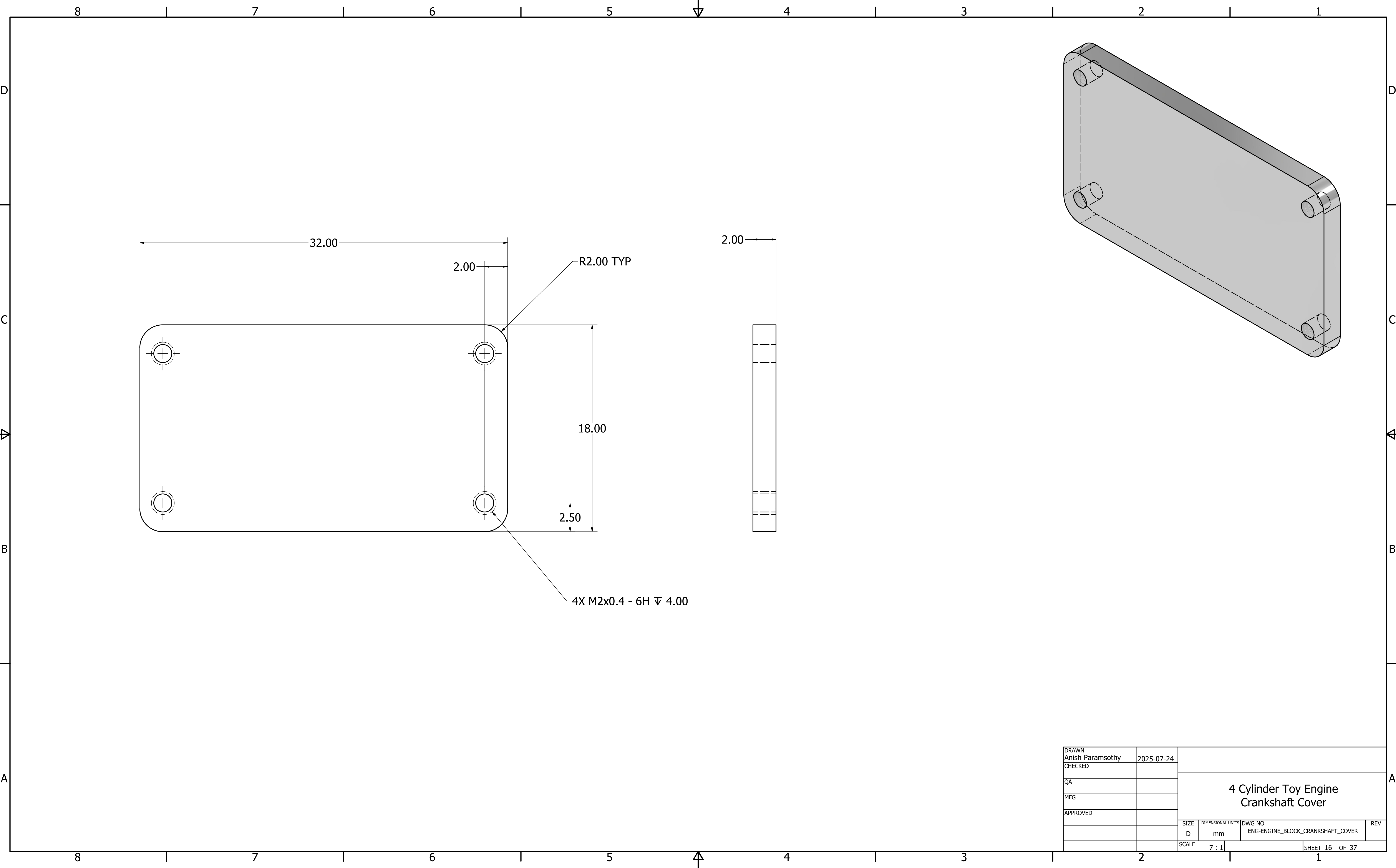


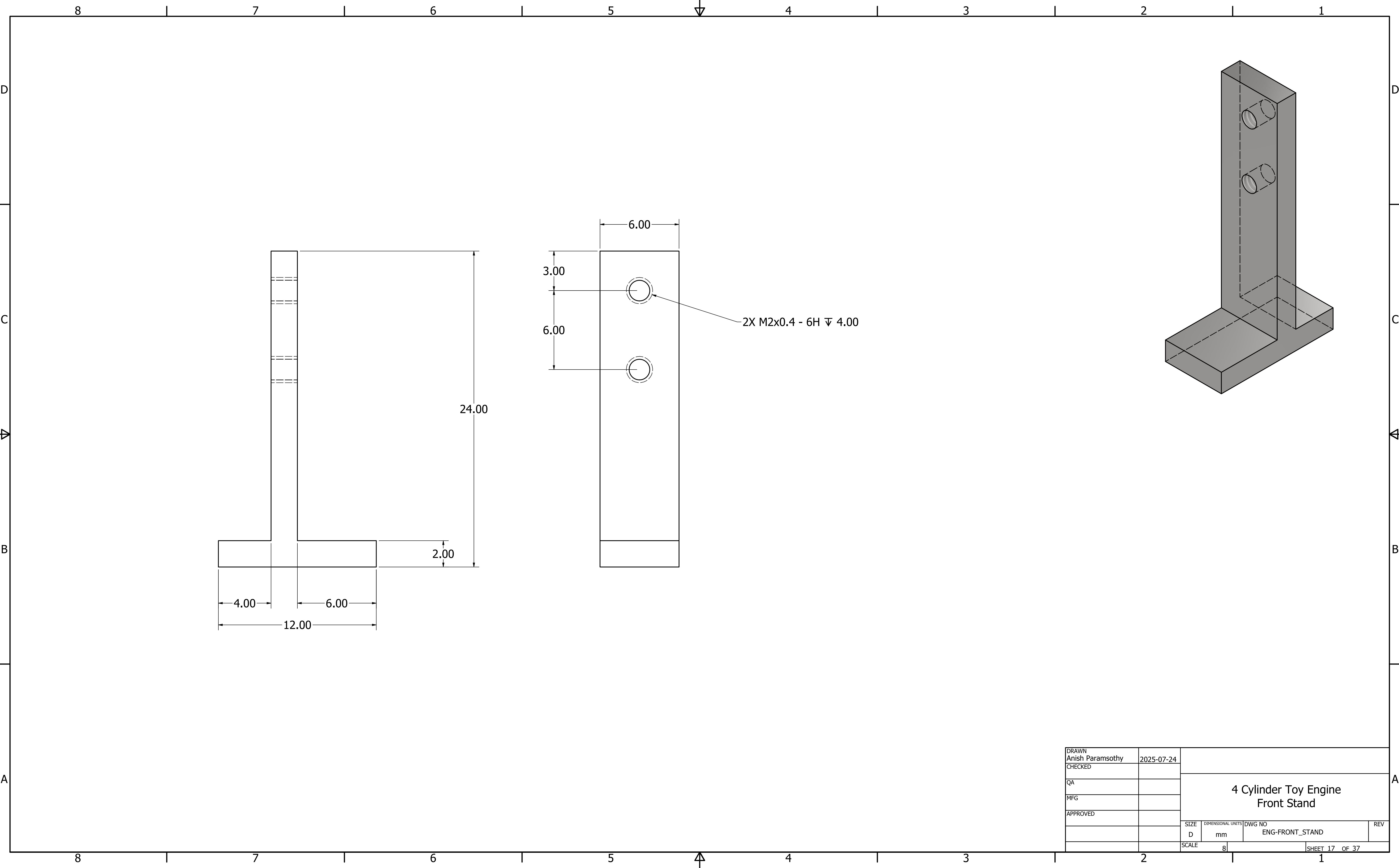
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Starter Motor Cover		
CHECKED					
QA			4 Cylinder Toy Engine Starter Motor Cover		
MFG					
APPROVED					
SIZE	D	DIMENSIONAL UNITS	mm	DWG NO	ENG-FLYWHEEL_STARTER_MOTOR_COVER
SCALE	10: 1			REV	
				SHEET 13 OF 37	



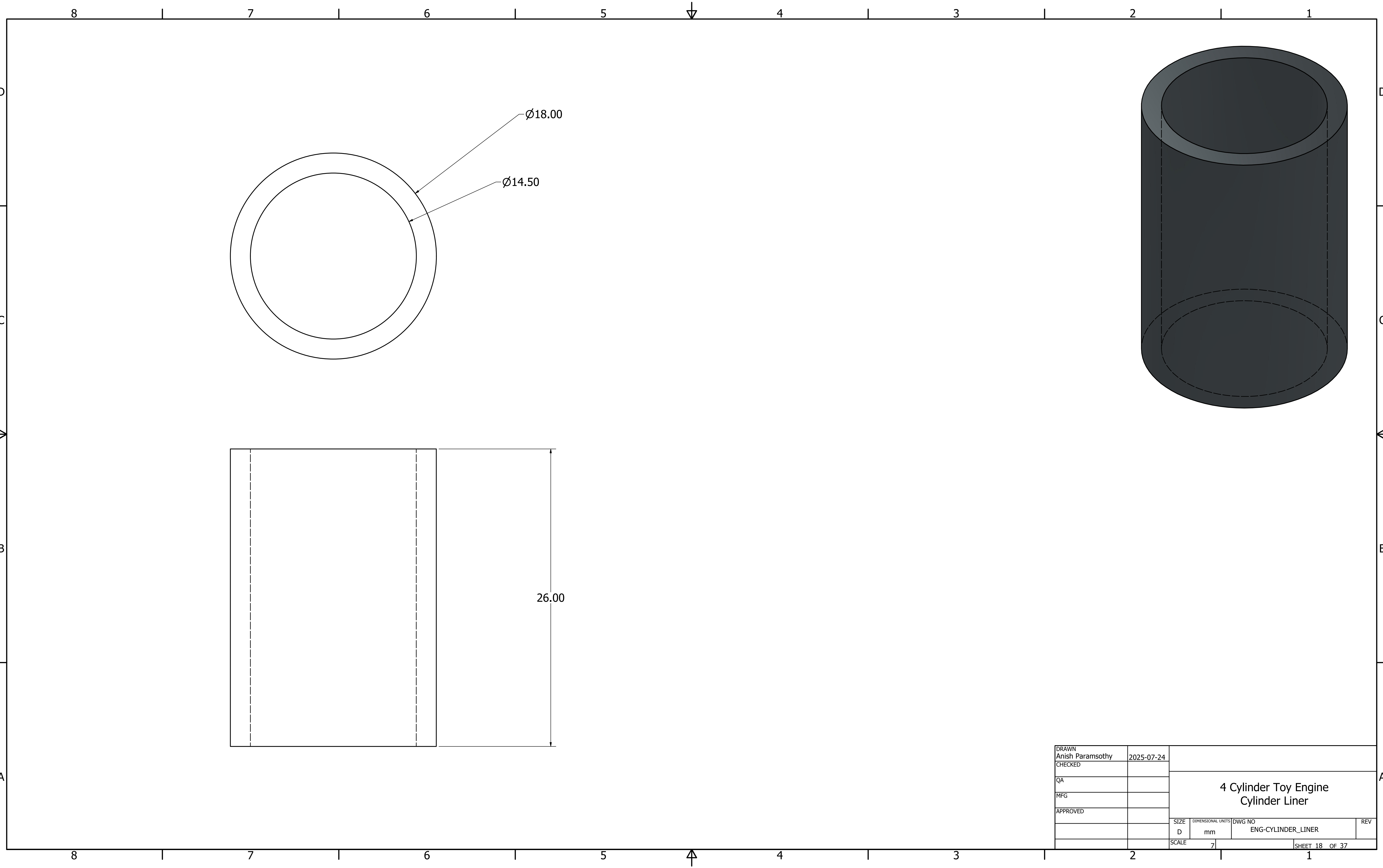


DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Intake Manifold		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-INTAKE_MANIFOLD
			SCALE	2: 1	SHEET 15 OF 37

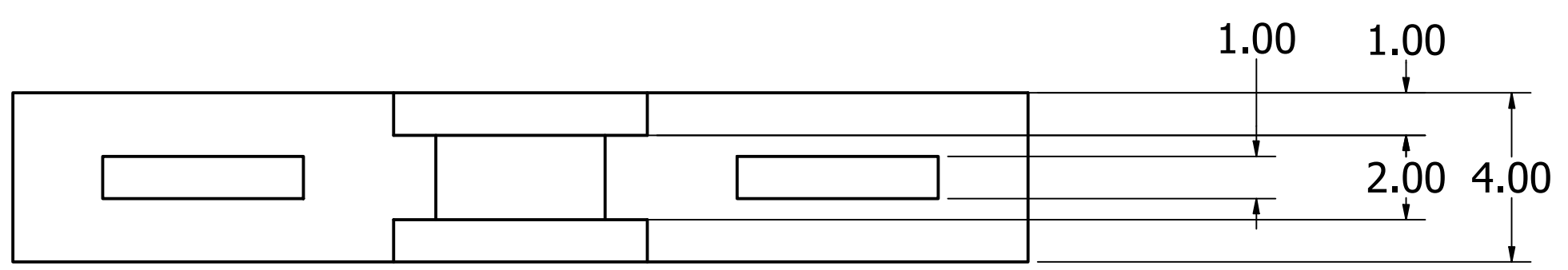
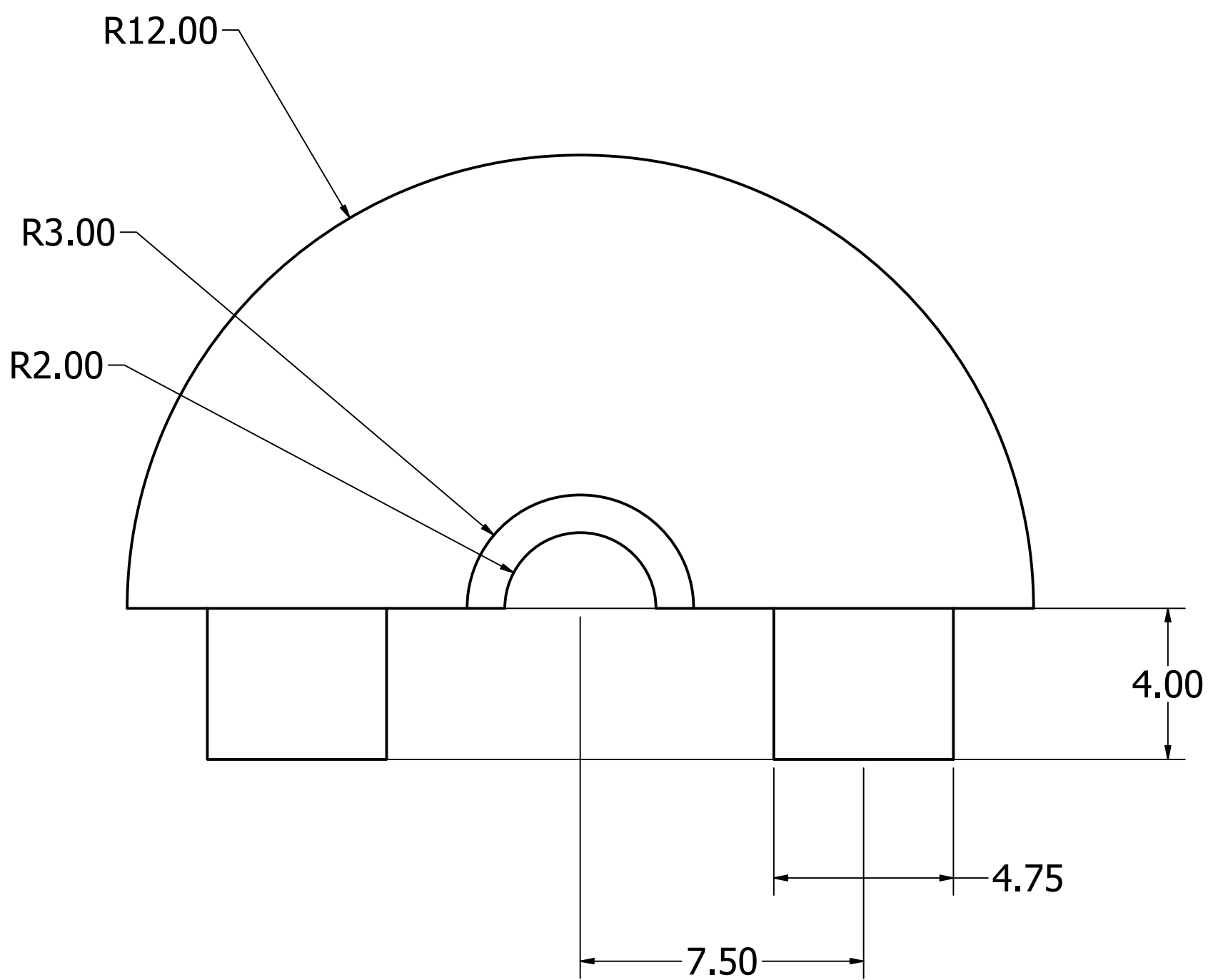
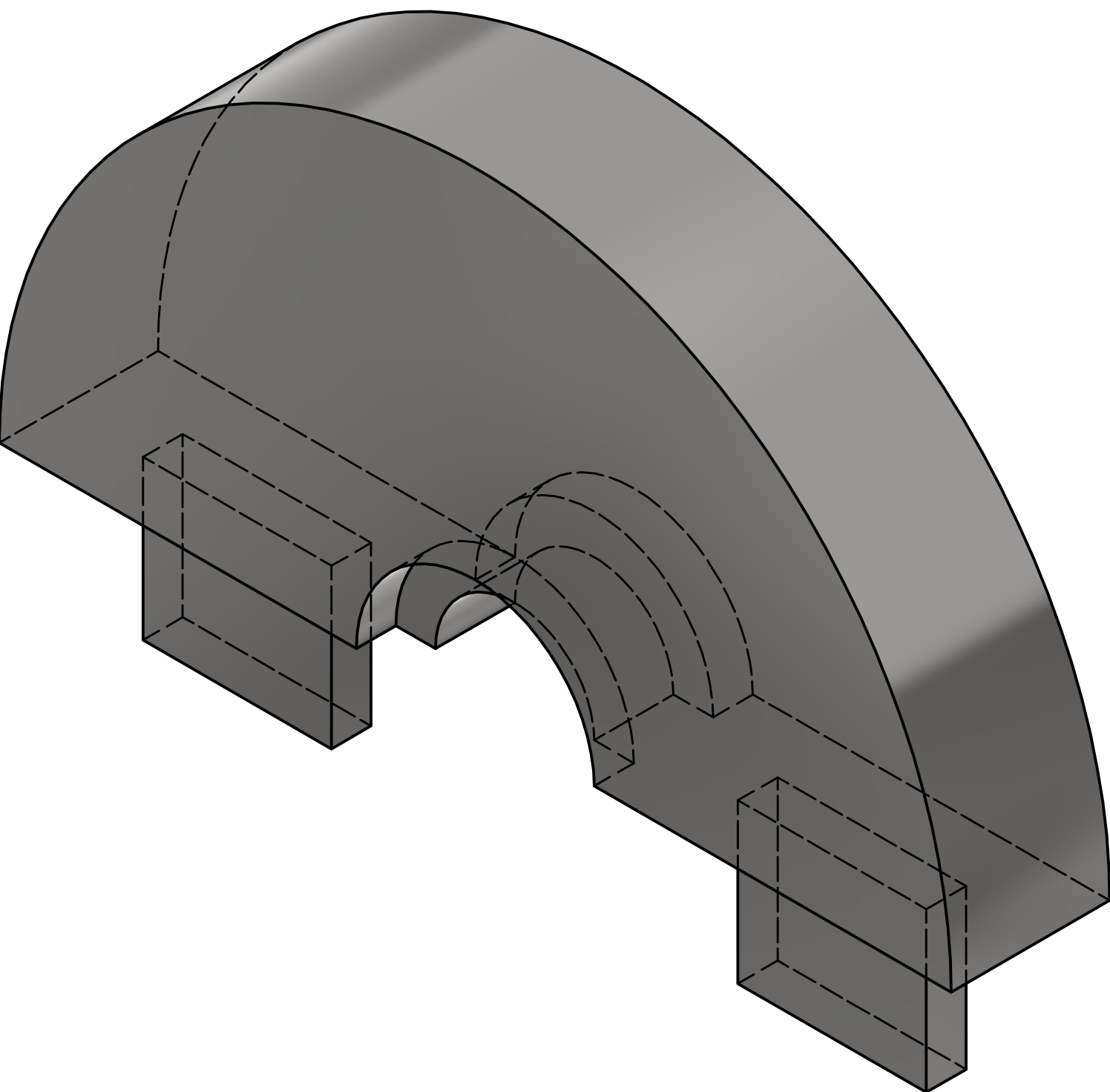




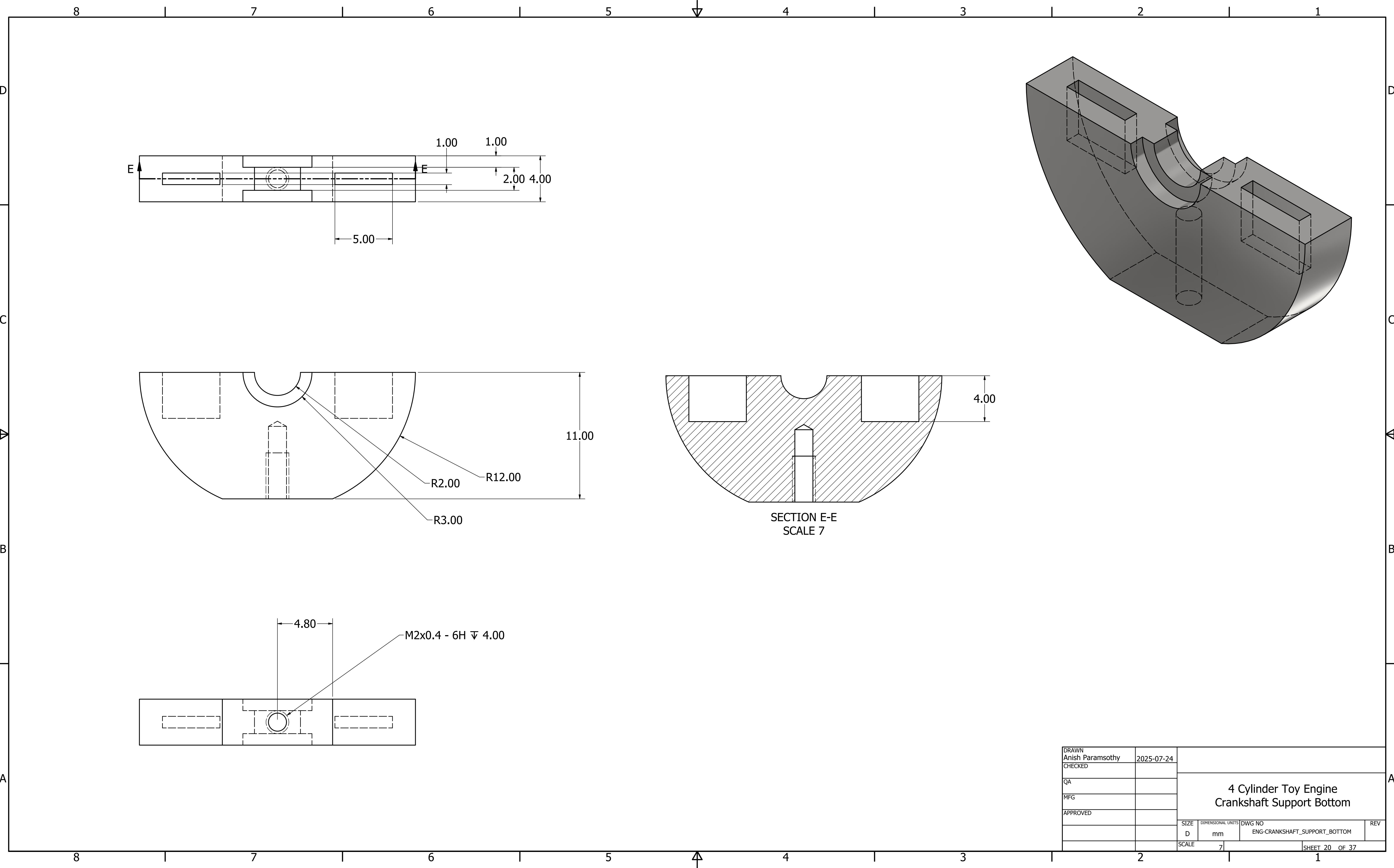
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Front Stand		
CHECKED					
QA					
MFG					
APPROVED					
SIZE	D	DIMENSIONAL UNITS	mm	DWG NO	ENG-FRONT_STAND
SCALE	8			SHEET 17 OF 37	

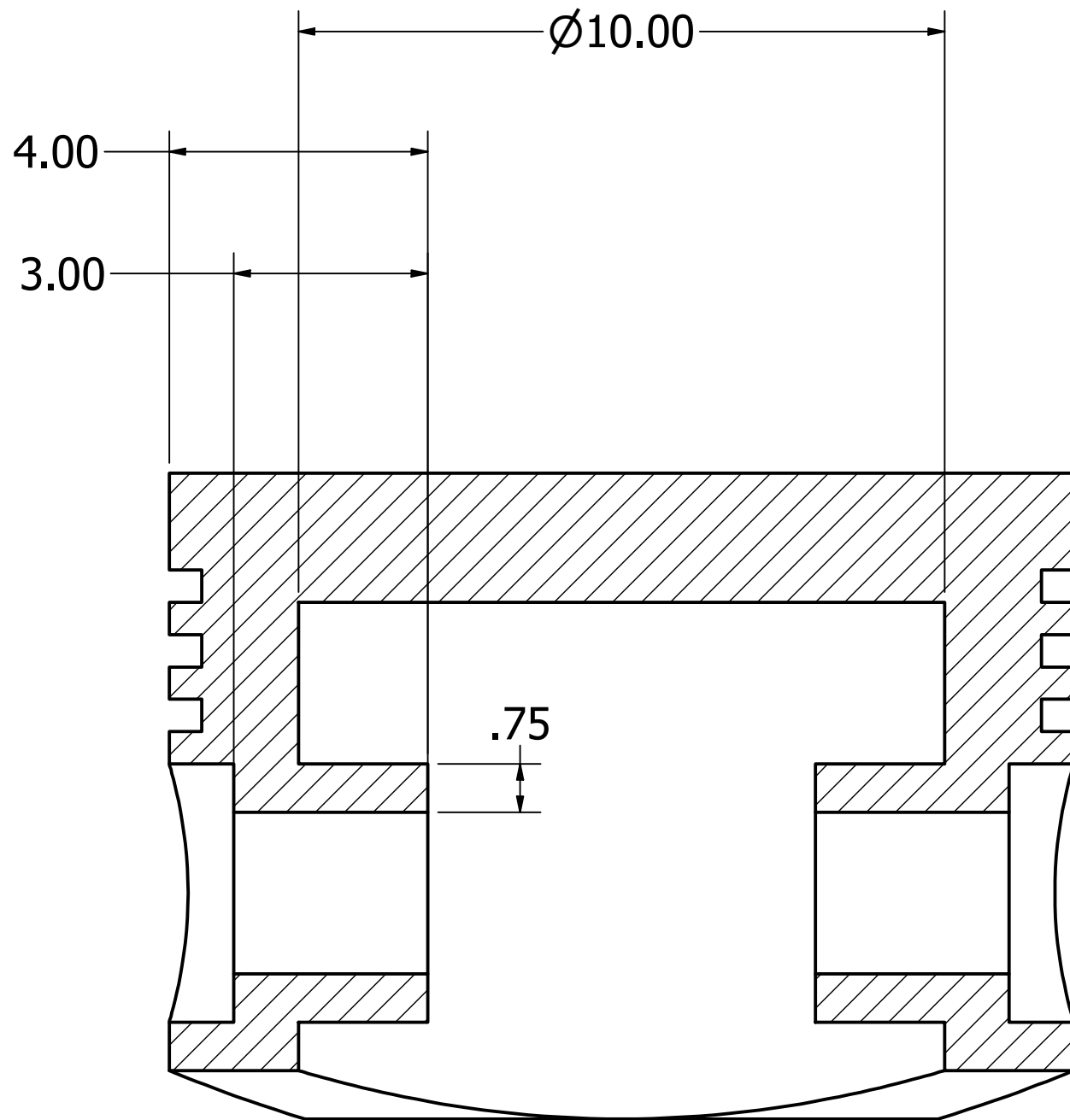
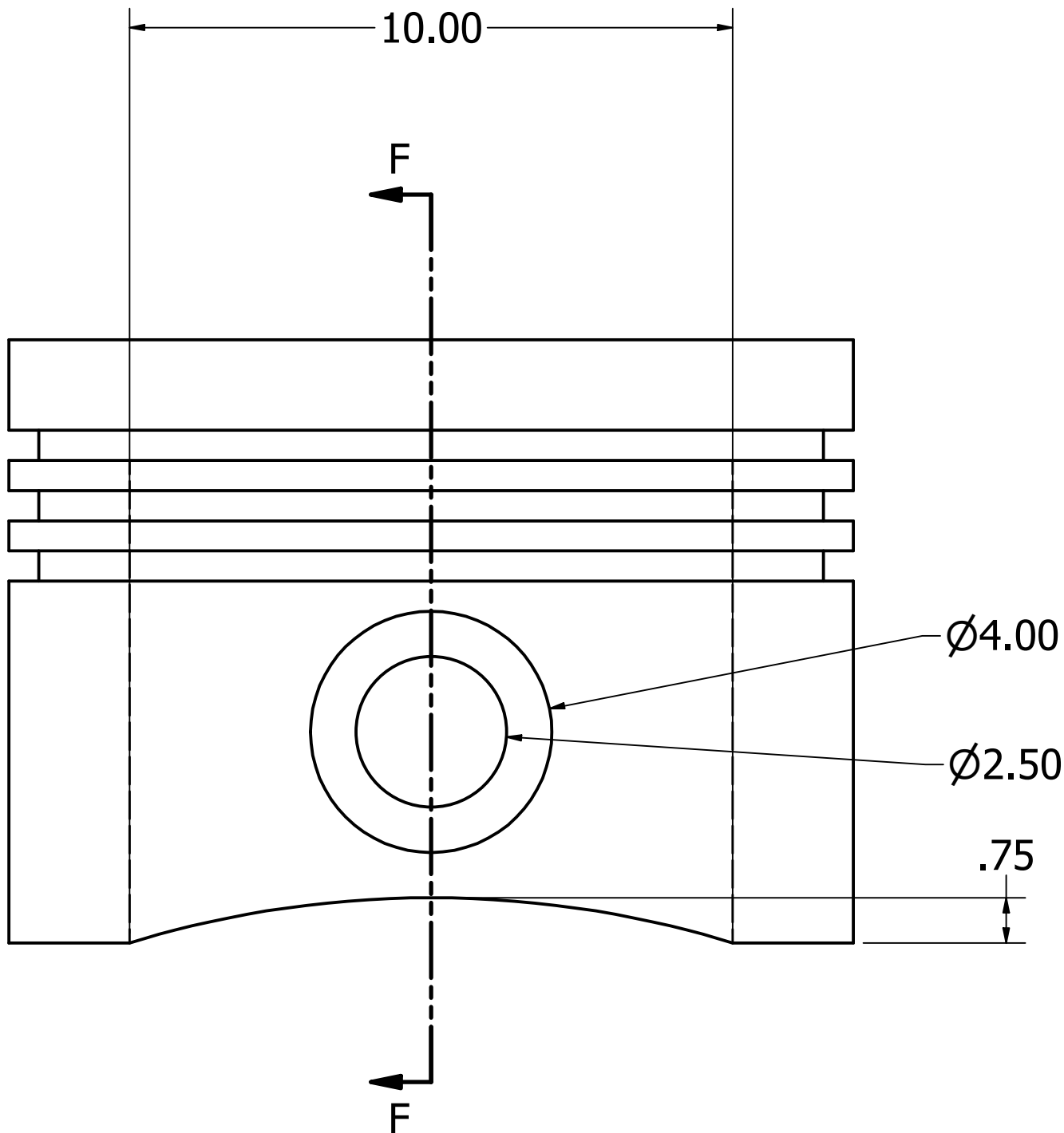
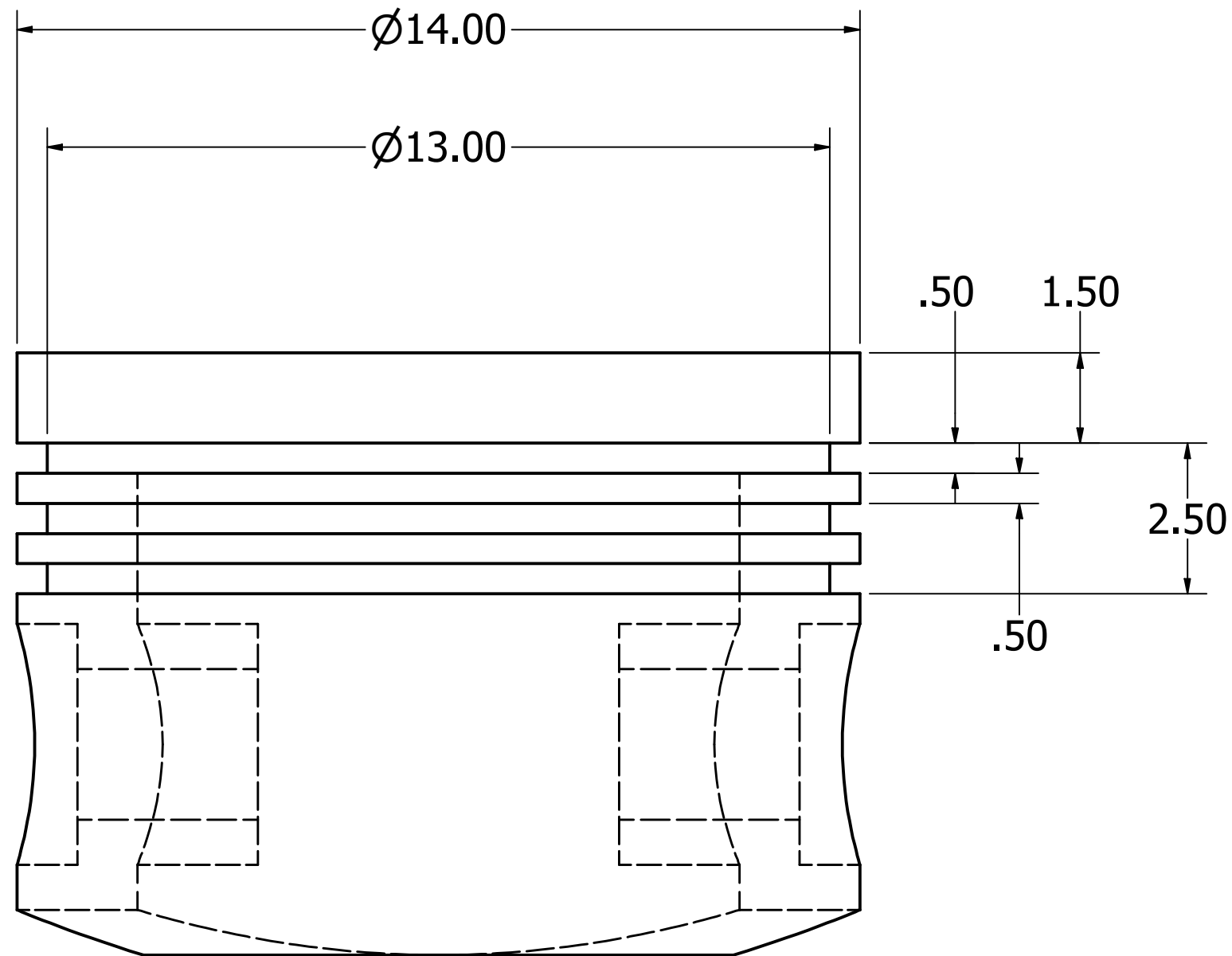
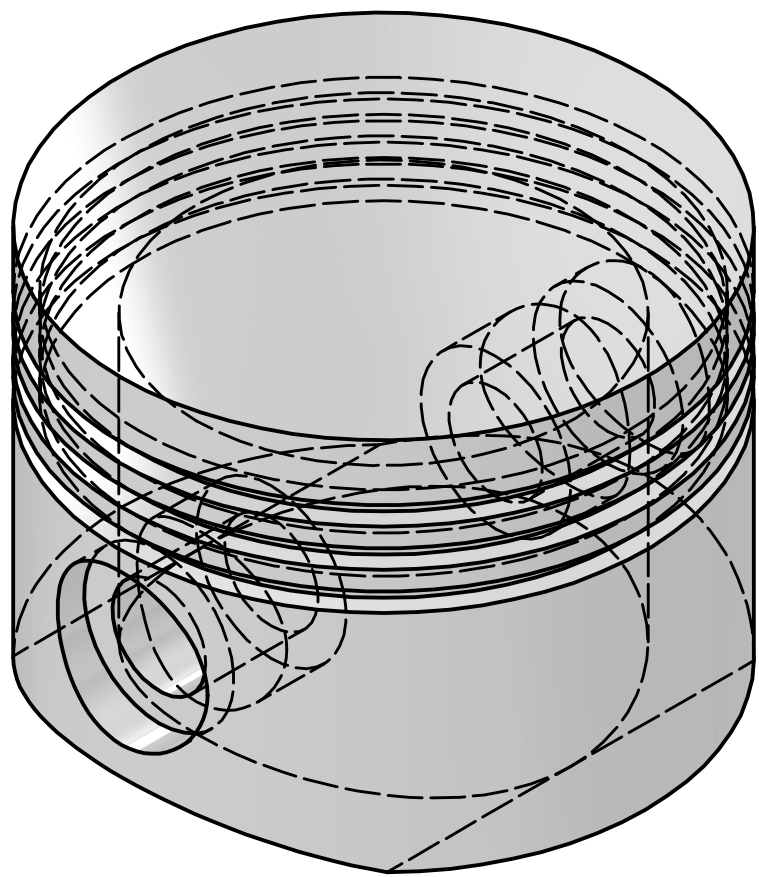


DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Cylinder Liner		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-CYLINDER_LINER
			SCALE	7	SHEET 18 OF 37



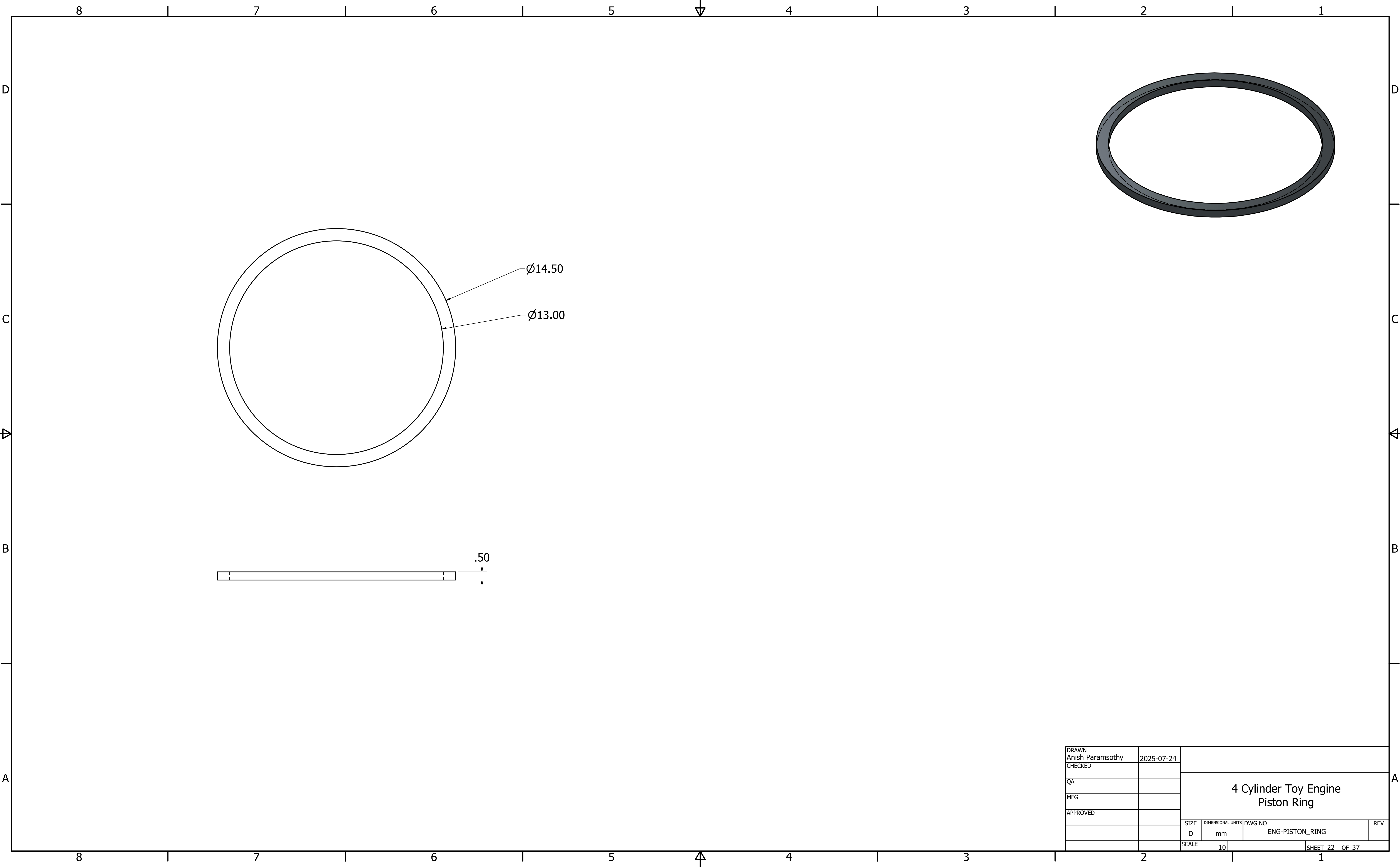
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Crankshaft Support Top		
CHECKED					
QA					
MFG					
APPROVED					
SIZE	D	DIMENSIONAL UNITS	mm	DWG NO	ENG-CRANKSHAFT_SUPPORT_TOP
SCALE	7	SHEET 19 OF 37			REV



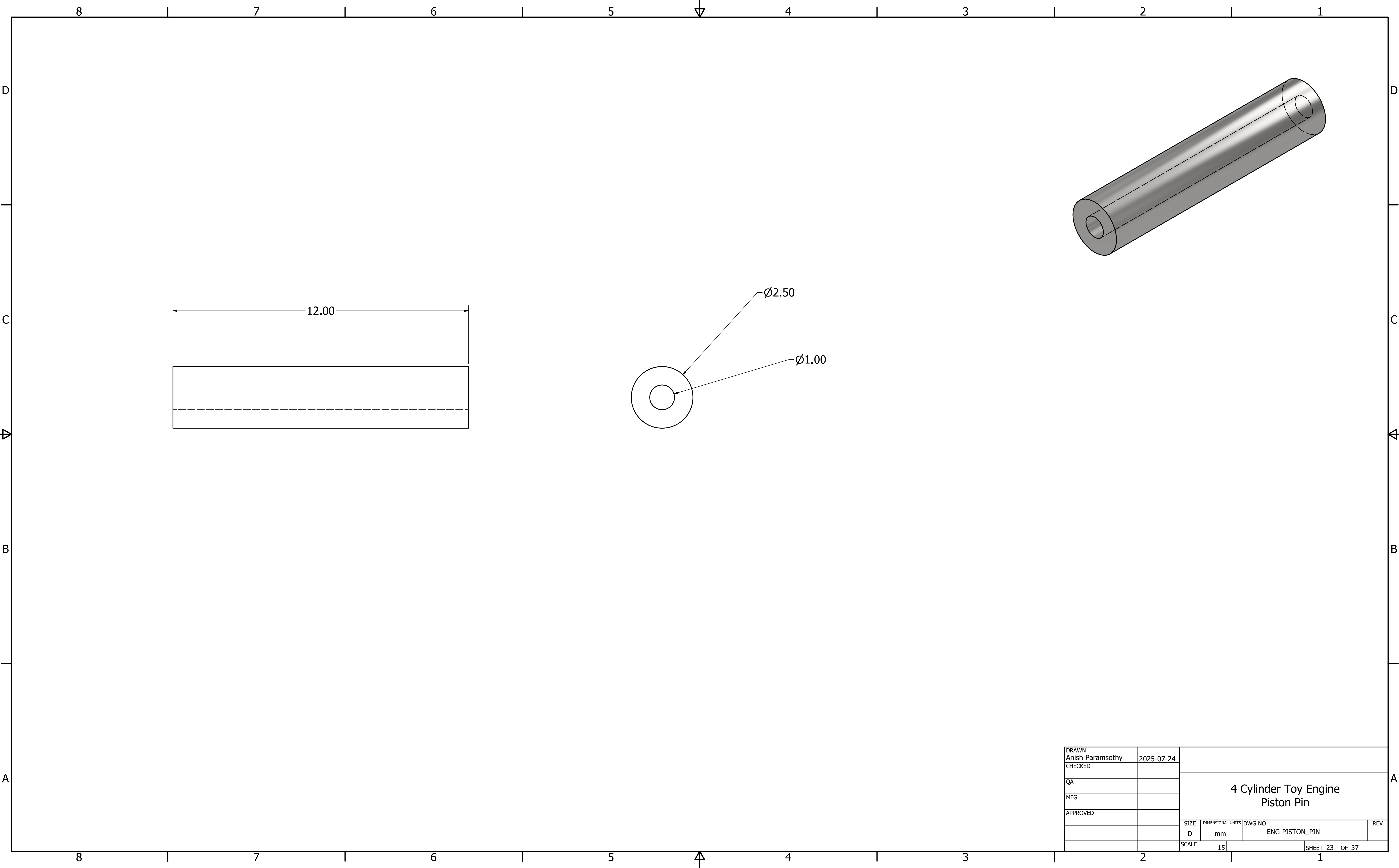


SECTION F-F
SCALE 10

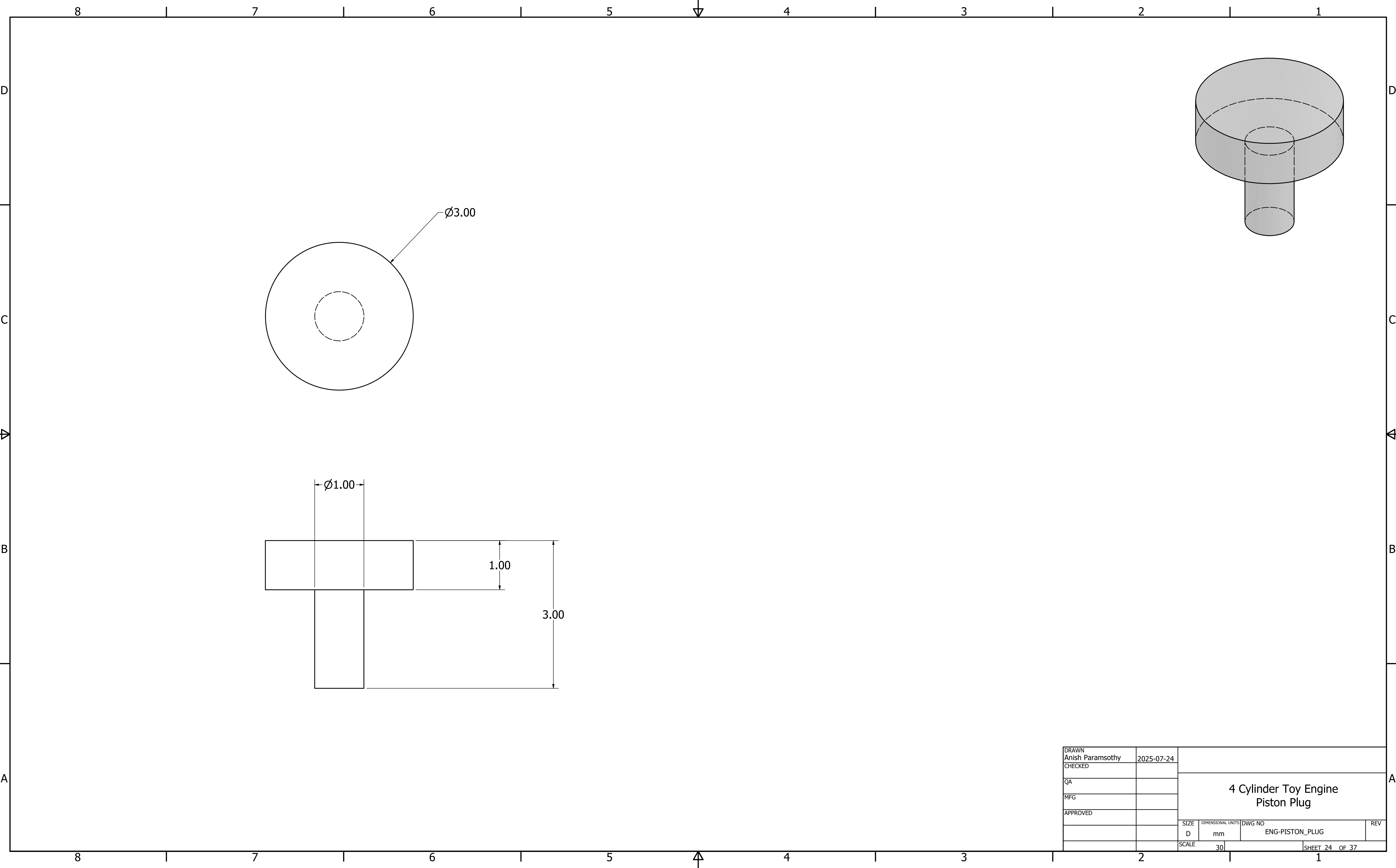
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Piston Head		
CHECKED					
QA					
MFG					
APPROVED					
SIZE	D	DIMENSIONAL UNITS	mm	DWG NO	ENG-PISTON_HEAD
SCALE	10	SHEET 21 OF 37			REV



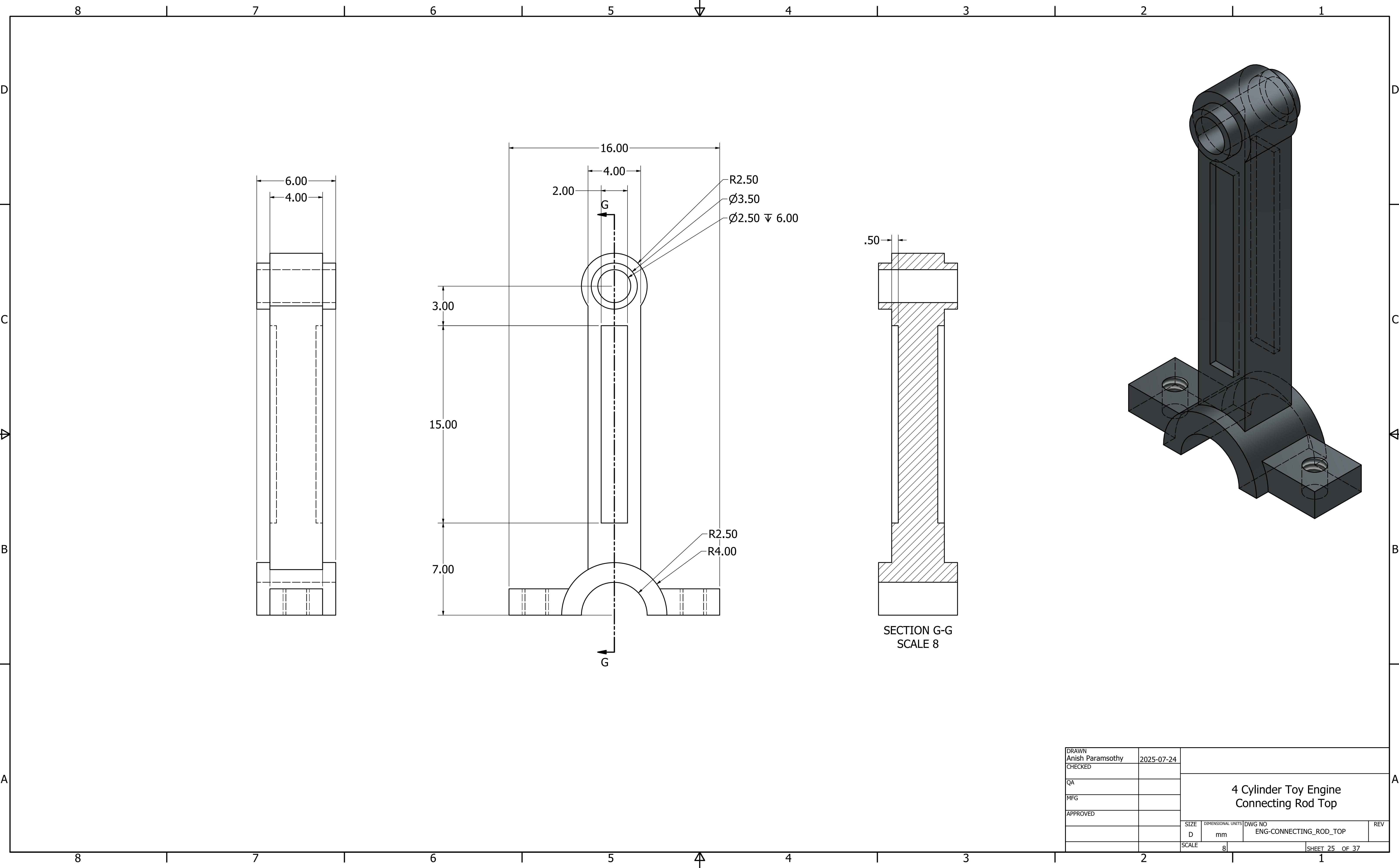
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Piston Ring		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-PISTON_RING
			SCALE	10	SHEET 22 OF 37



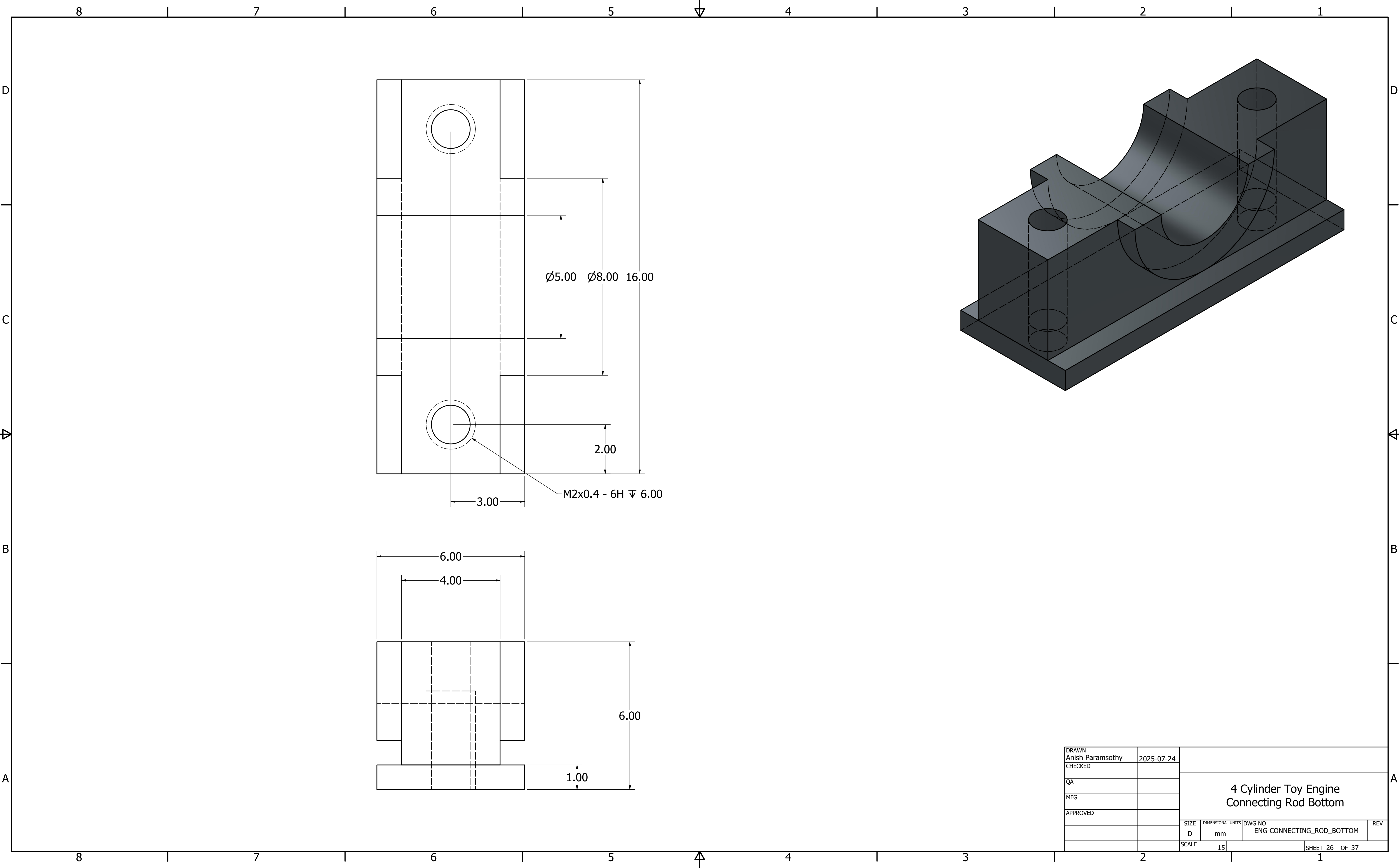
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Piston Pin		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-PISTON_PIN
			SCALE	15	SHEET 23 OF 37

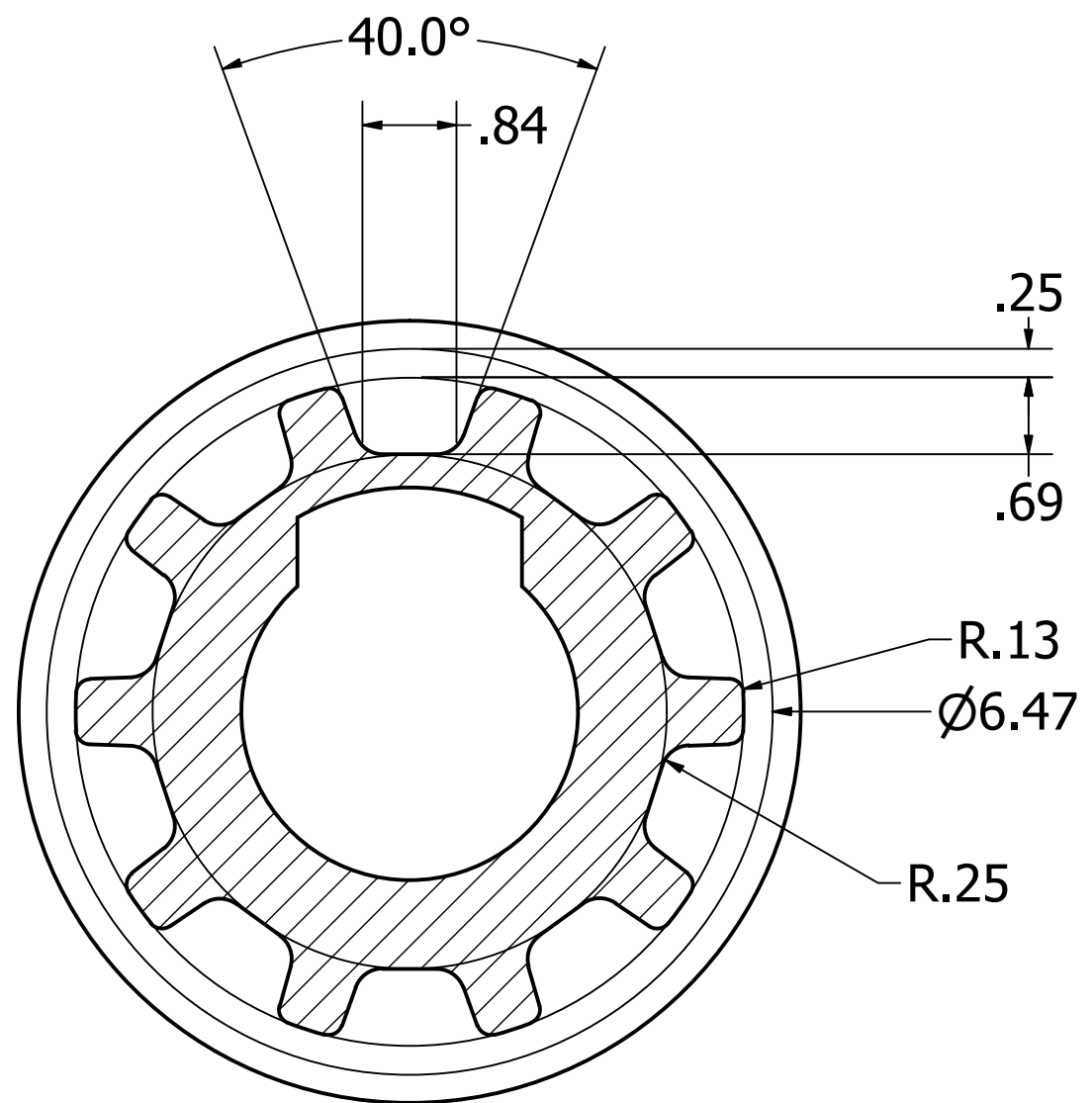


DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Piston Plug			
CHECKED						
QA						
MFG						
APPROVED						
SIZE	D	mm	DWG NO	ENG-PISTON_PLUG	REV	
SCALE	30		SHEET 24	OF 37		

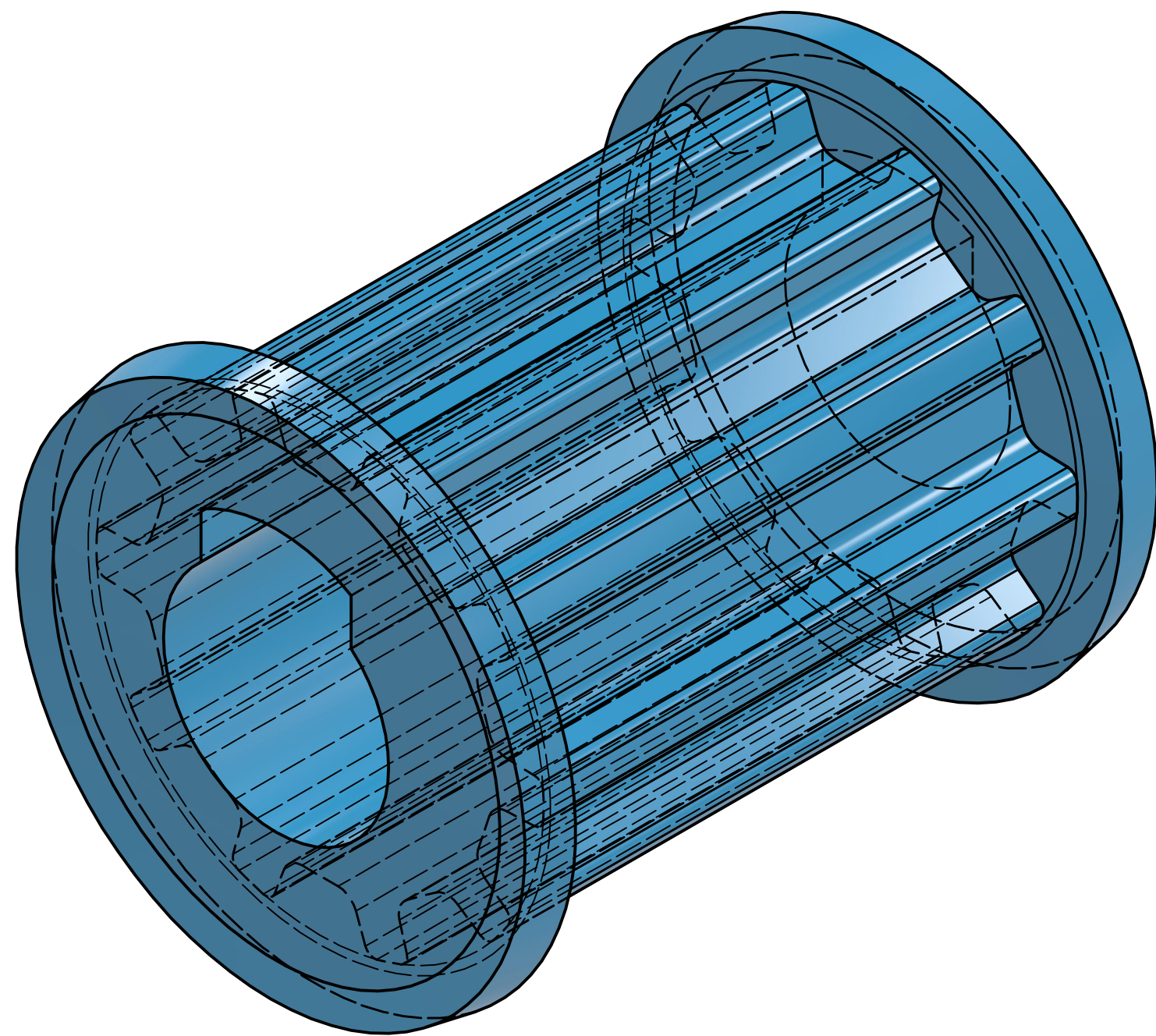
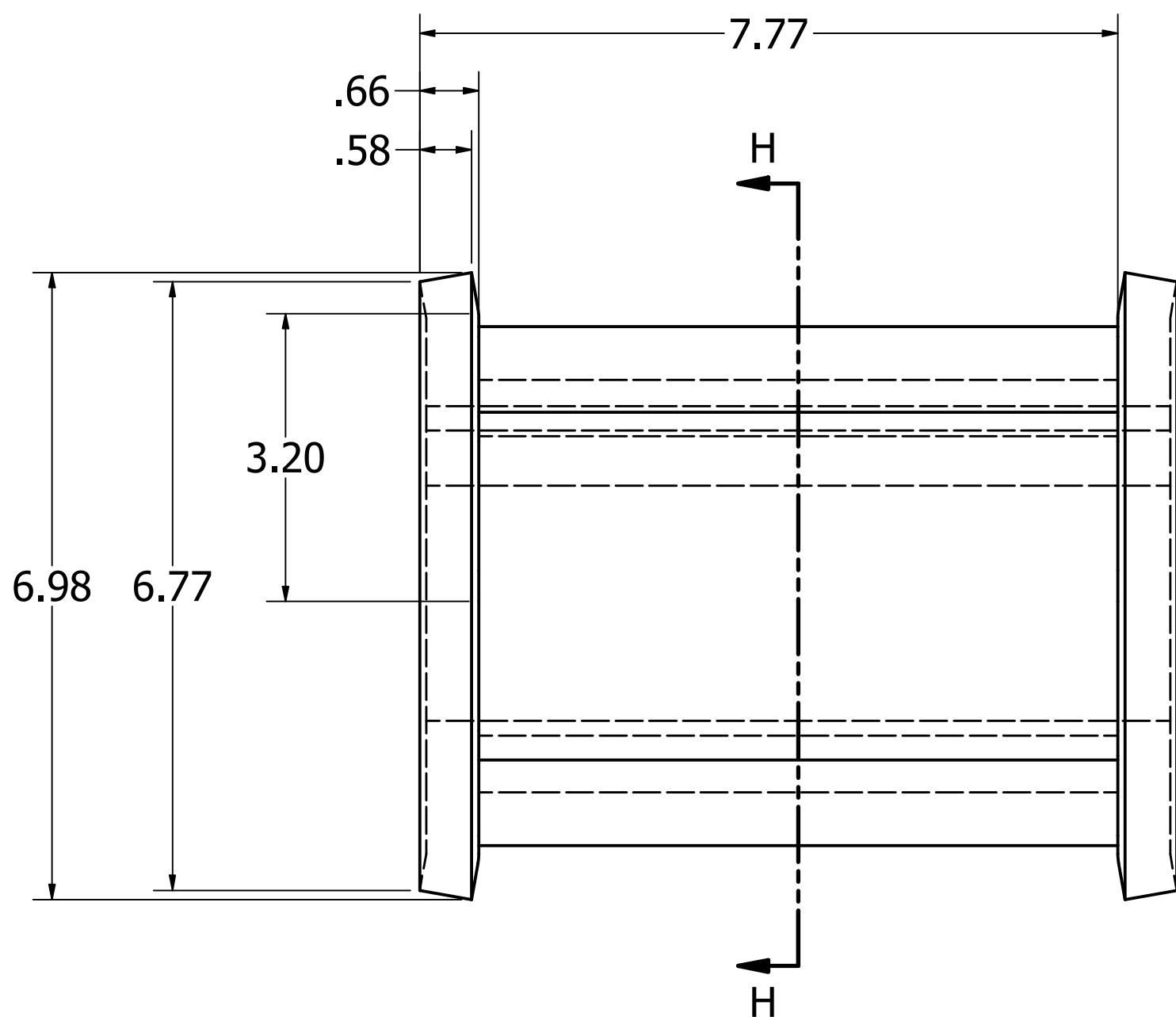
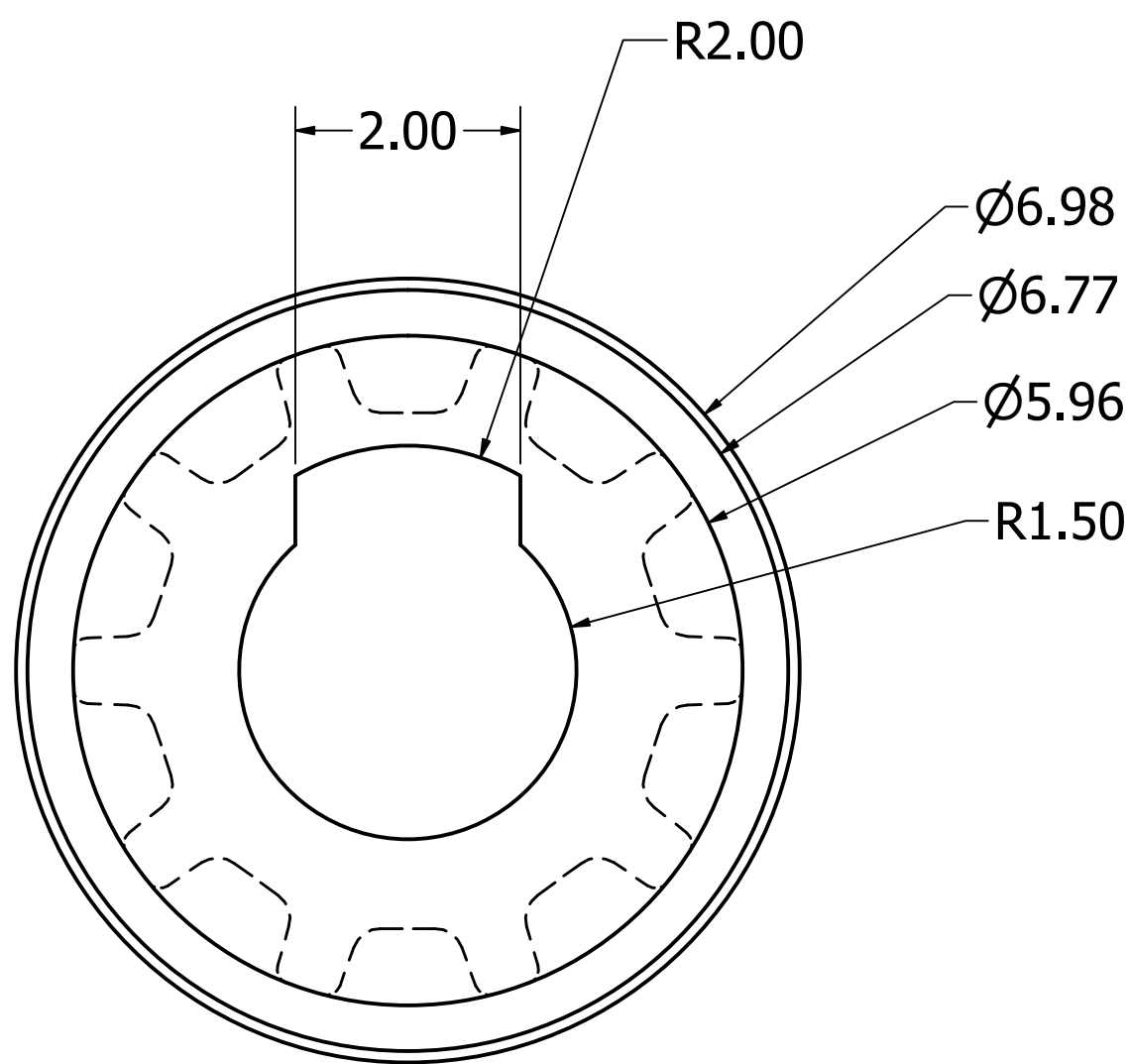


DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Connecting Rod Top		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE D	DIMENSIONAL UNITS mm	DWG NO ENG-CONNECTING_ROD_TOP
			SCALE 8		REV
					SHEET 25 OF 37

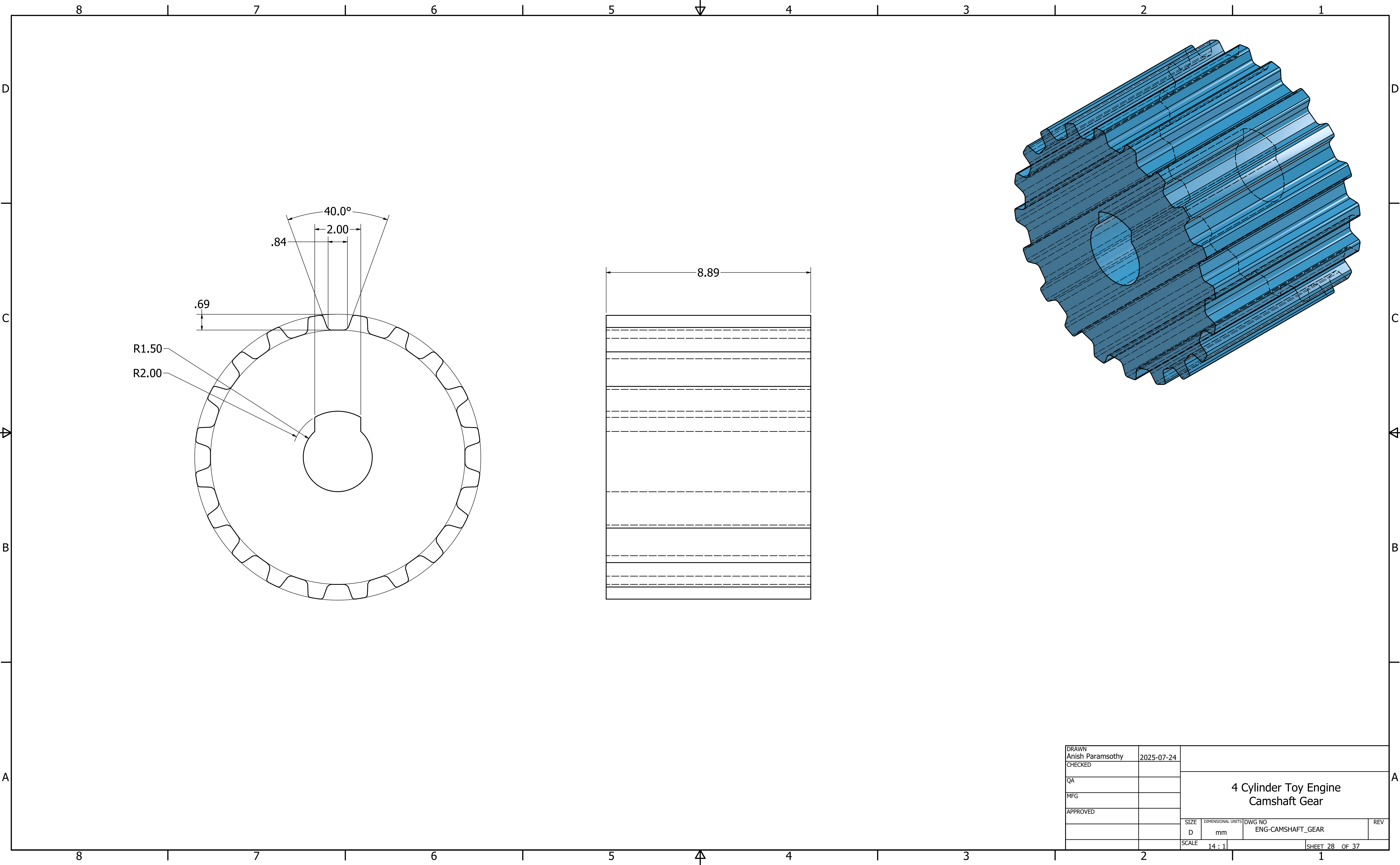




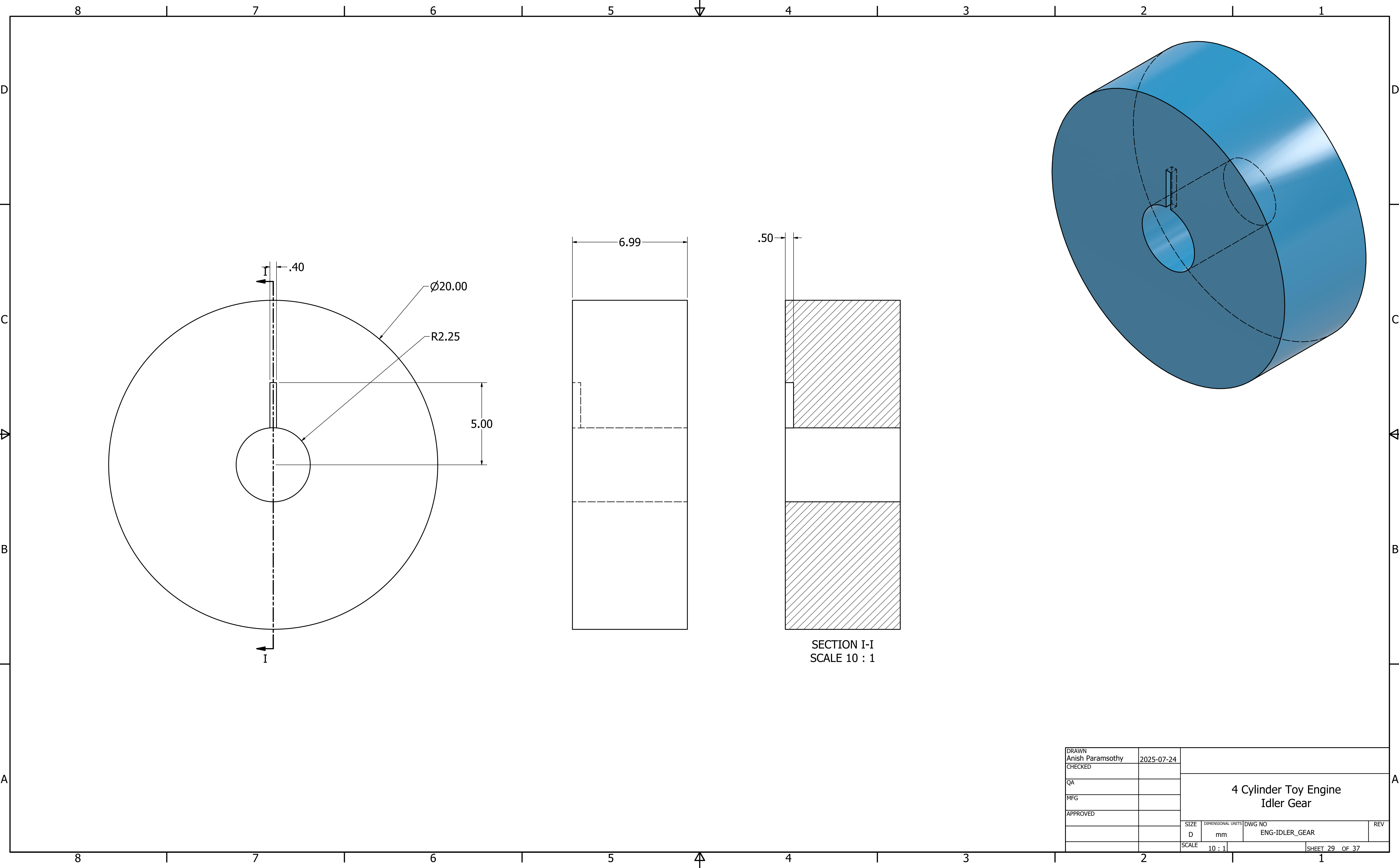
SECTION H-H
SCALE 15: 1



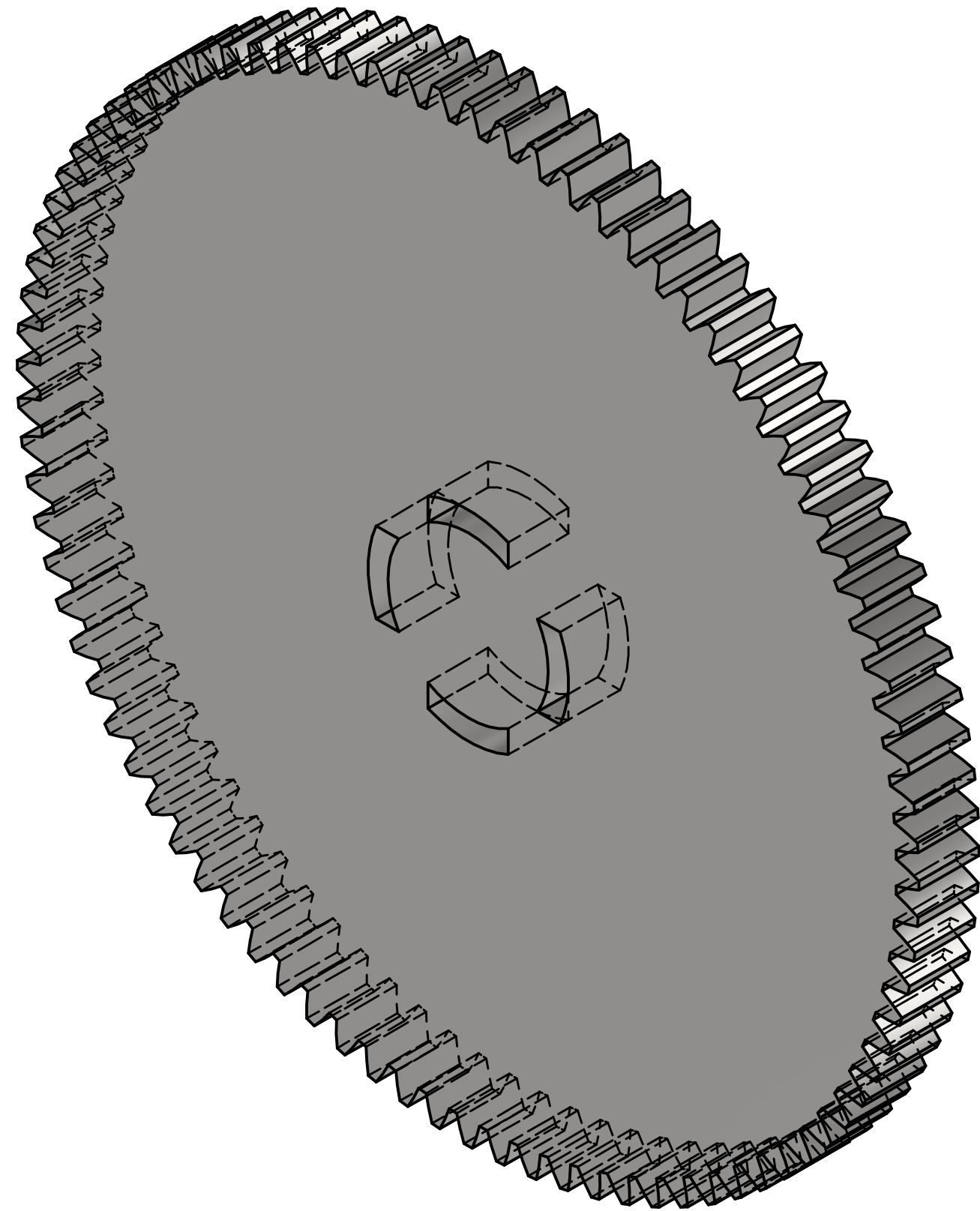
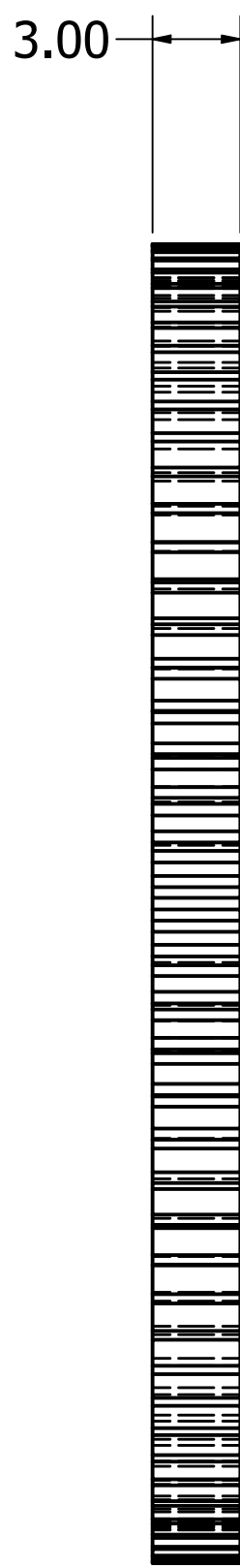
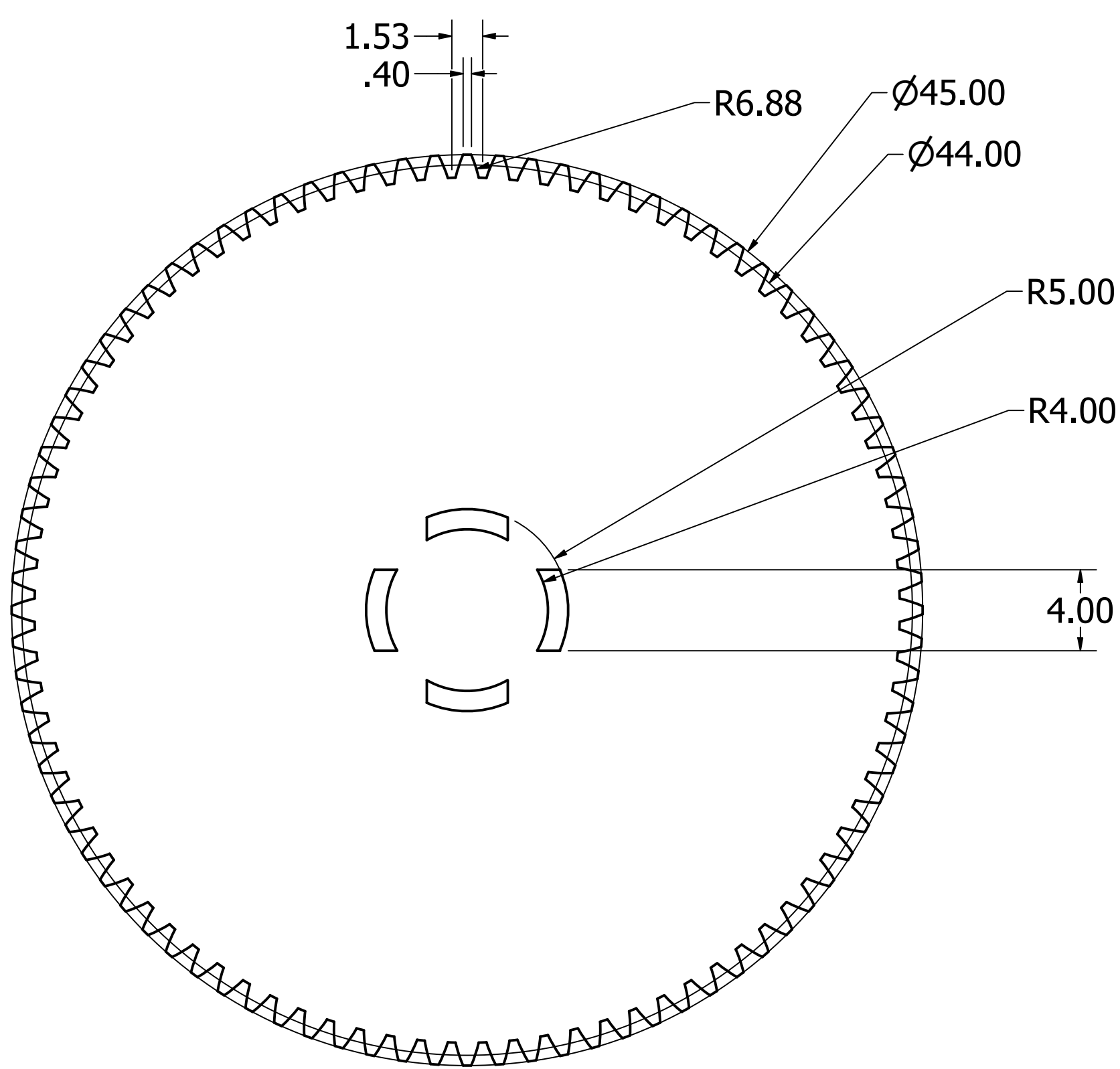
DRAWN Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Crankshaft Gear		
CHECKED				
QA				
MFG				
APPROVED				
		SIZE D	DIMENSIONAL UNITS mm	DWG NO ENG-CRANKSHAFT_GEAR
		SCALE 15: 1		REV
				SHEET 27 OF 37



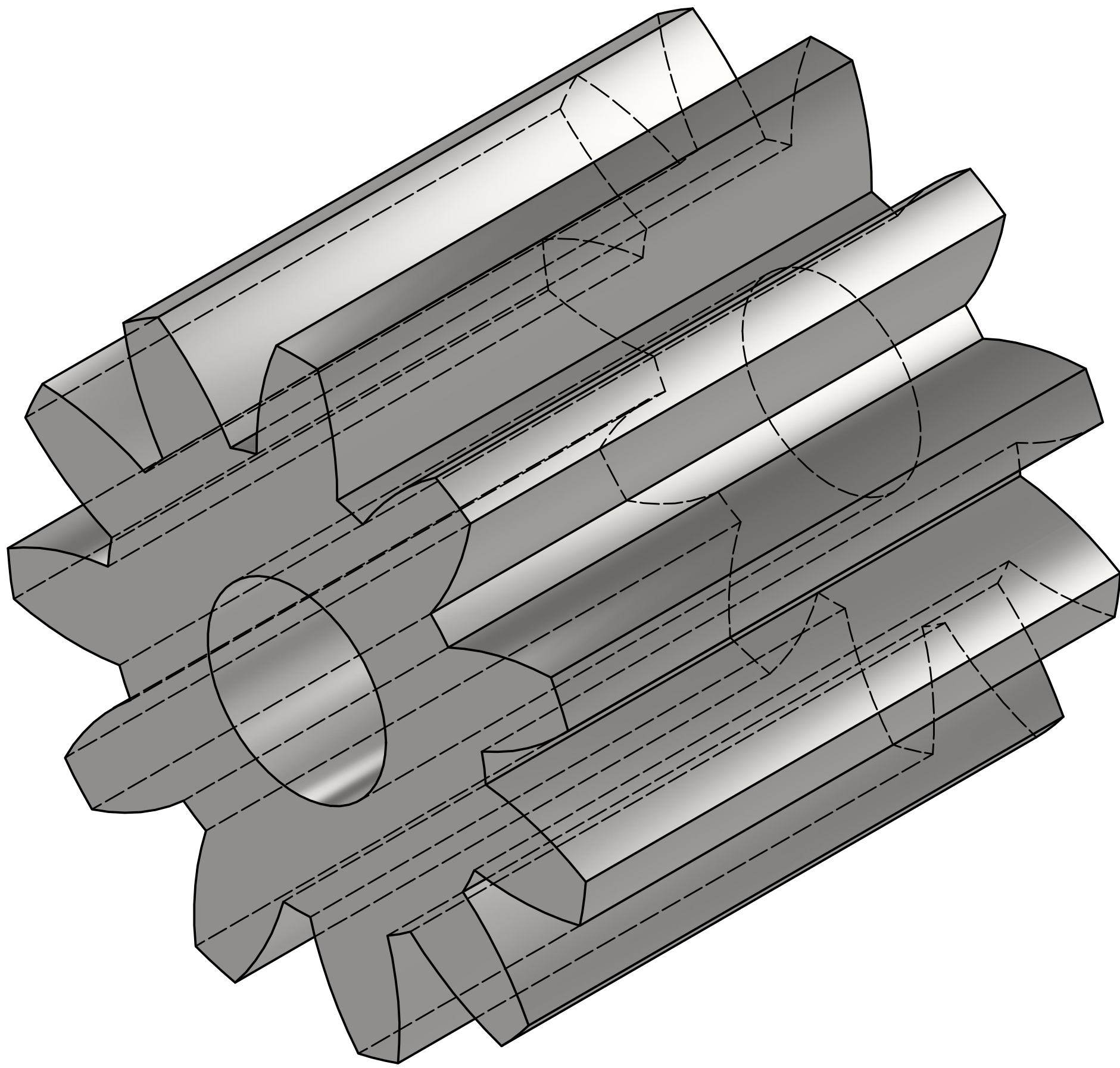
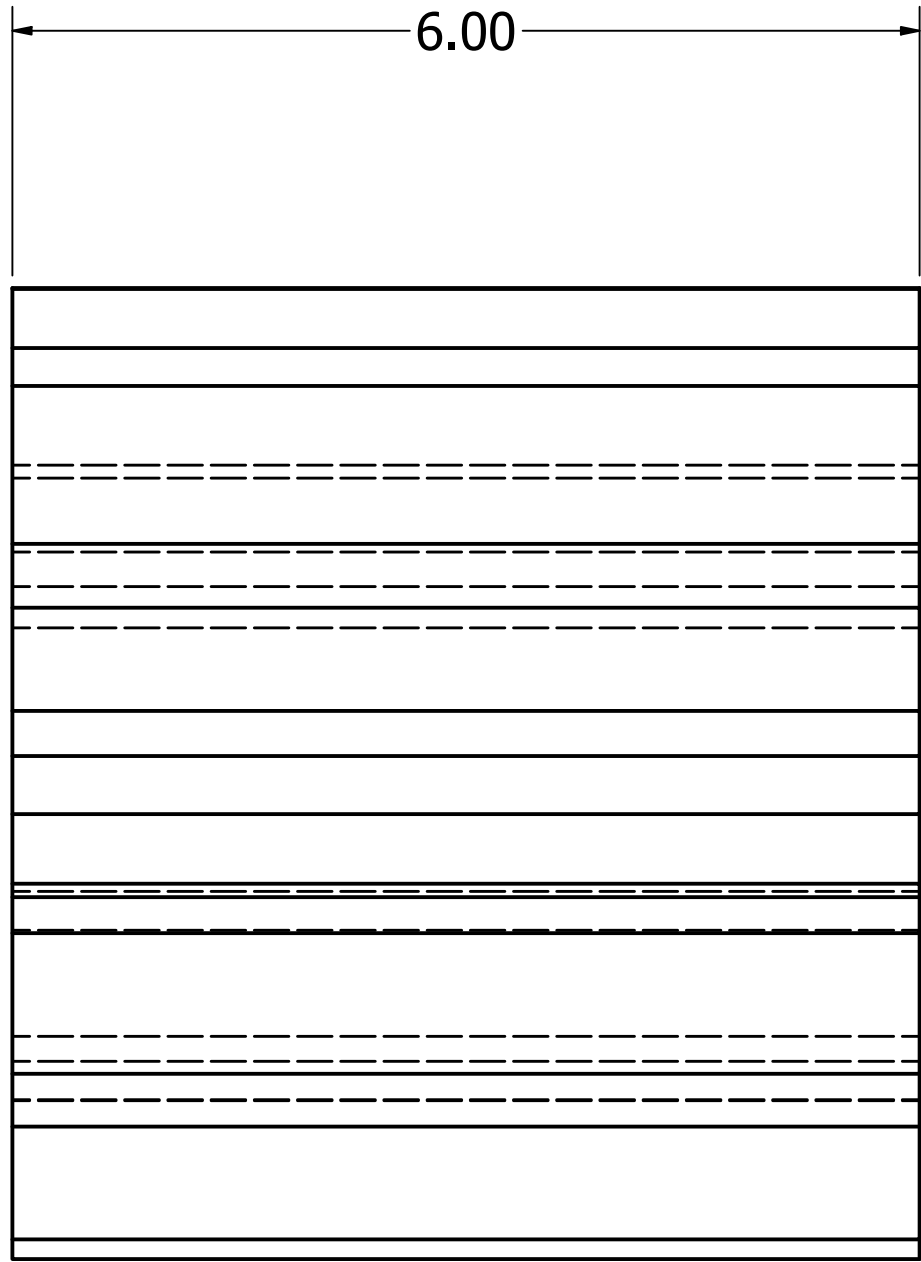
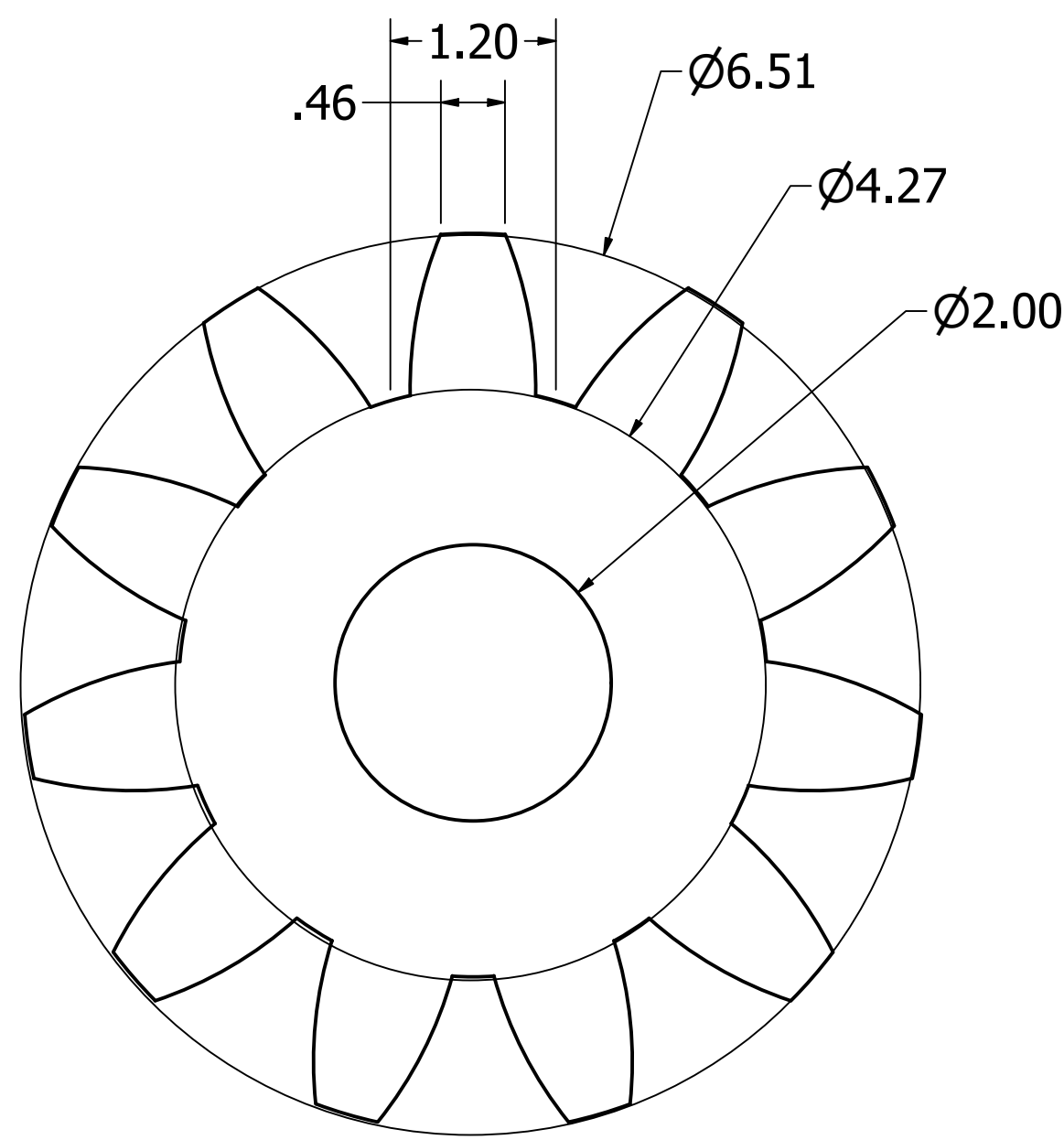
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Camshaft Gear		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-CAMSHAFT_GEAR
			SCALE	14 : 1	SHEET 28 OF 37



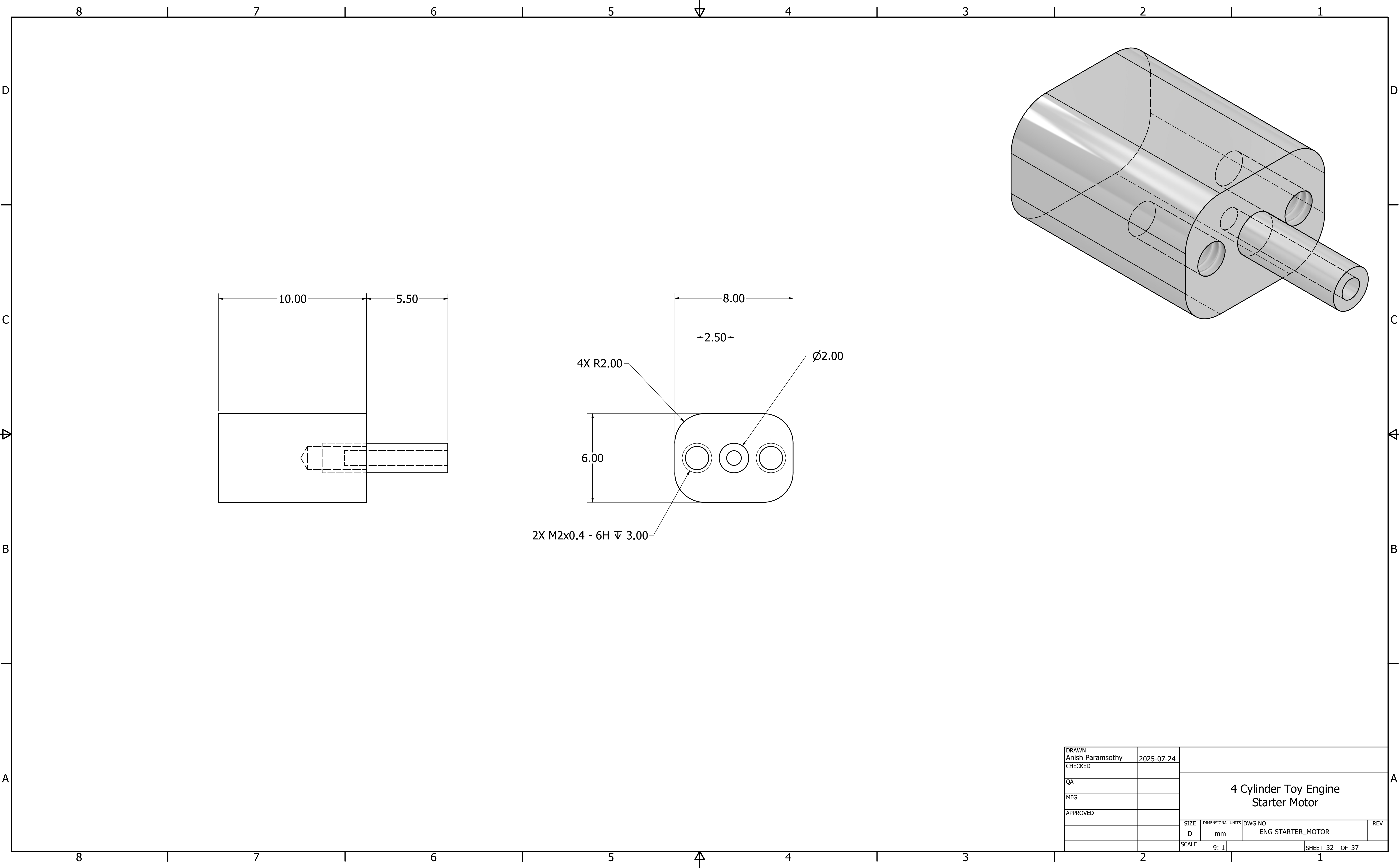
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Idler Gear			
CHECKED						
QA						
MFG						
APPROVED						
			SIZE	DIMENSIONAL UNITS	DWG NO	REV
			D	mm	ENG-IDLER_GEAR	
			SCALE	10 : 1	SHEET 29 OF 37	



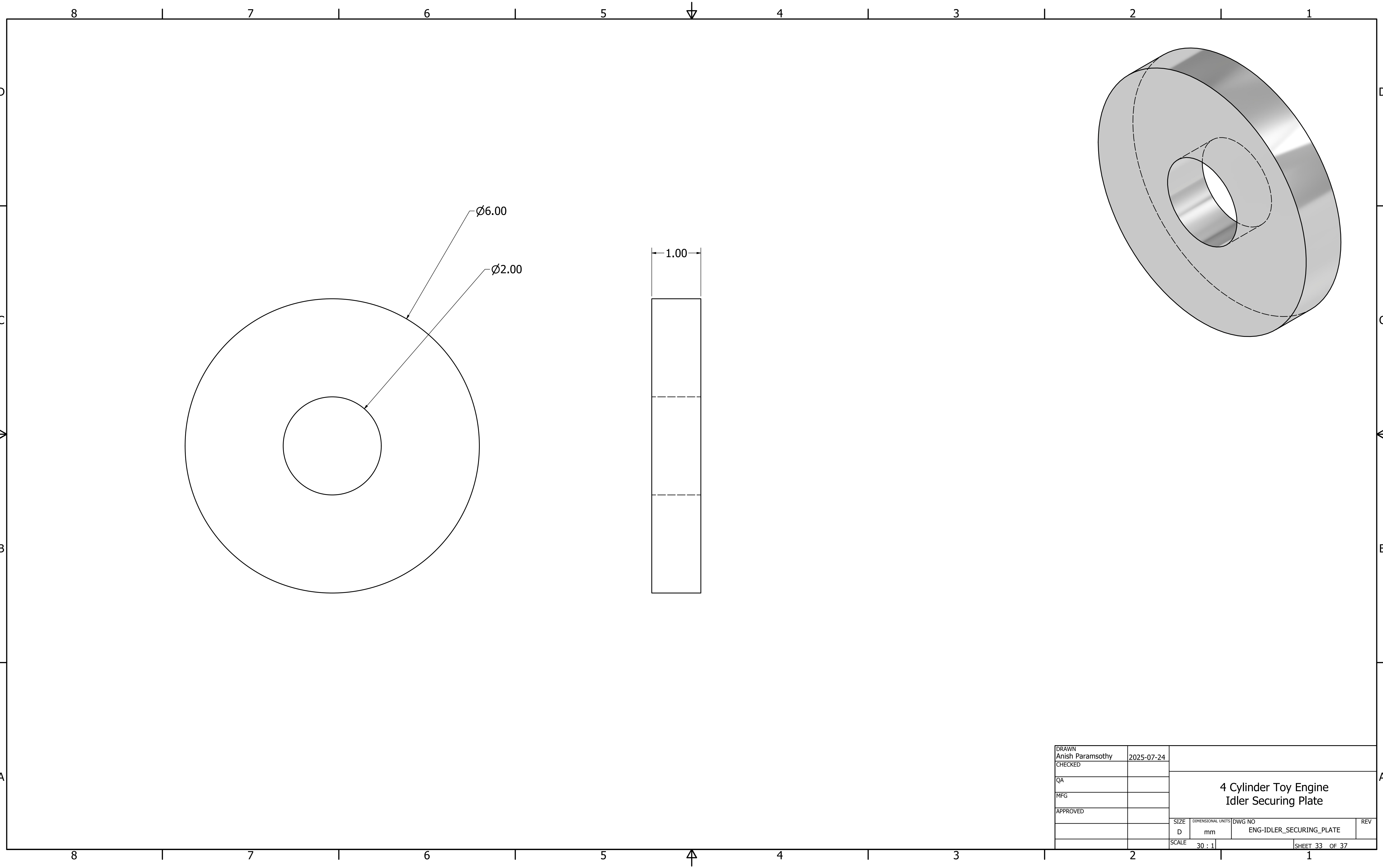
DRAWN Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Flywheel Gear		
CHECKED				
QA				
MFG				
APPROVED		SIZE D	DIMENSIONAL UNITS mm	DWG NO ENG-FLYWHEEL_GEAR
		SCALE 4 : 1	SHEET 30 OF 37	



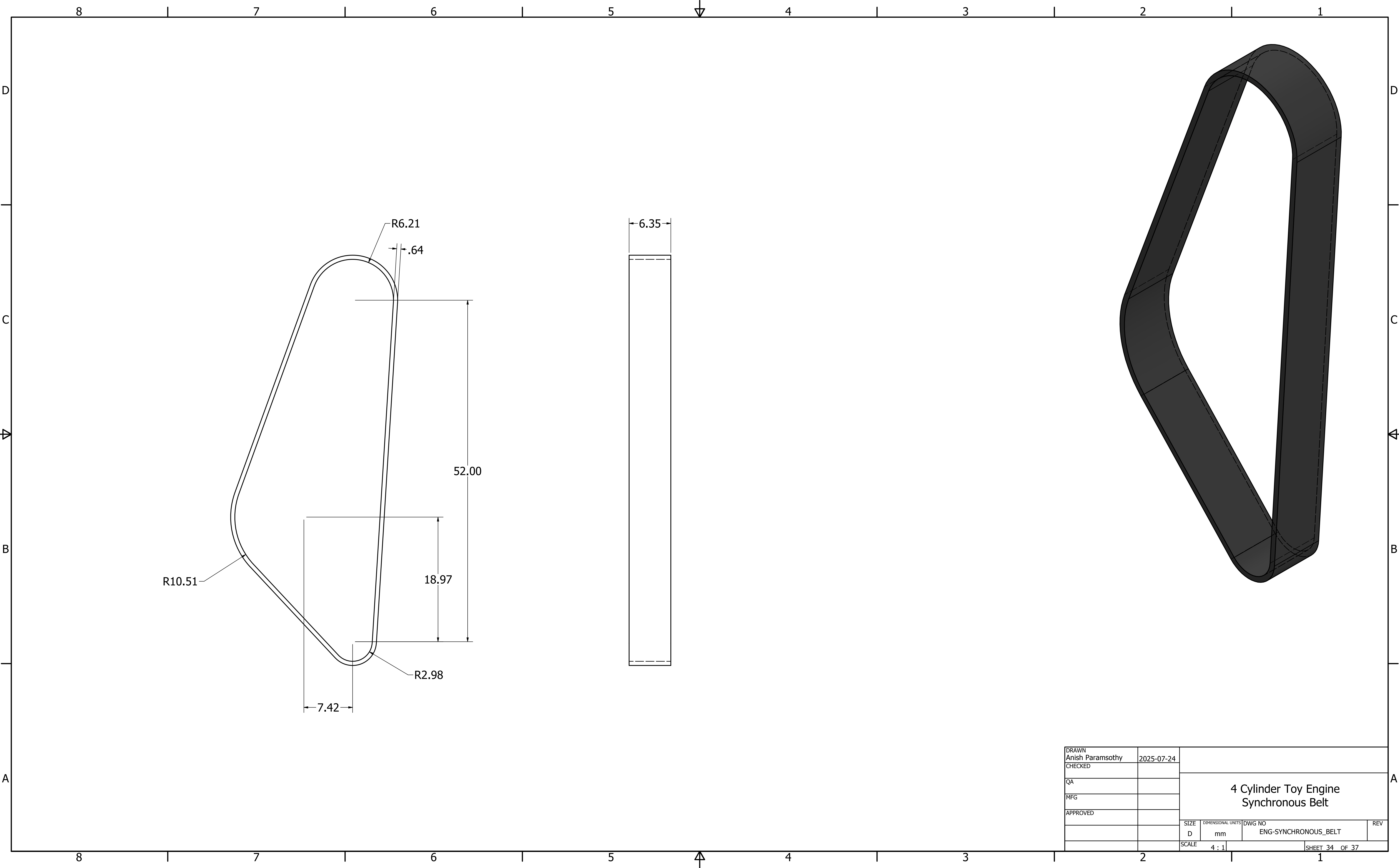
DRAWN Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Starter Motor Gear			
CHECKED					
QA					
MFG					
APPROVED					
		SIZE D	DIMENSIONAL UNITS mm	DWG NO ENG-STARTER_MOTOR_GEAR	REV
		SCALE 20 : 1			SHEET 31 OF 37



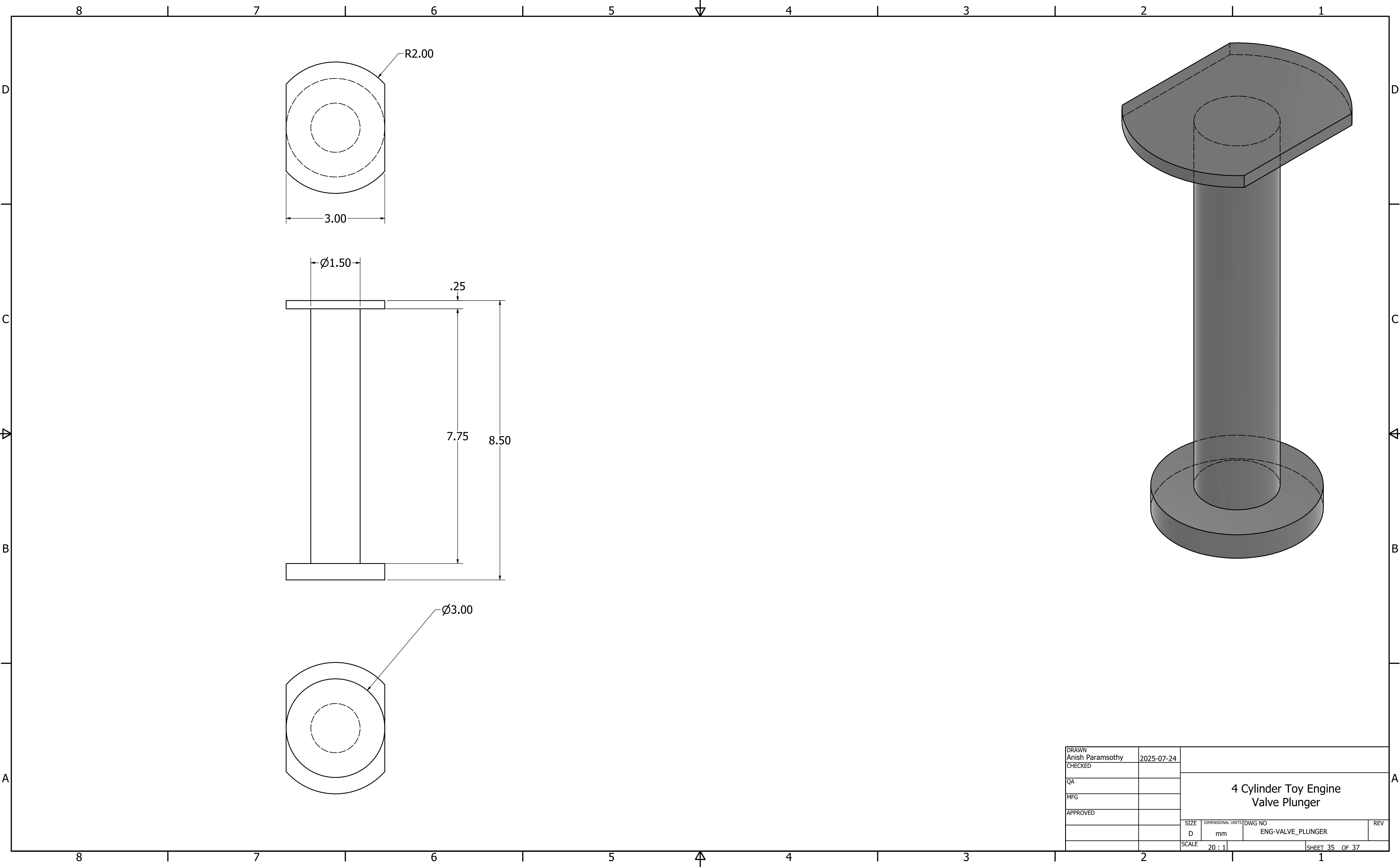
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Starter Motor		
CHECKED					
QA					
MFG					
APPROVED					
SIZE	D	DIMENSIONAL UNITS	mm	DWG NO	ENG-STARTER_MOTOR
SCALE	9: 1			SHEET 32 OF 37	



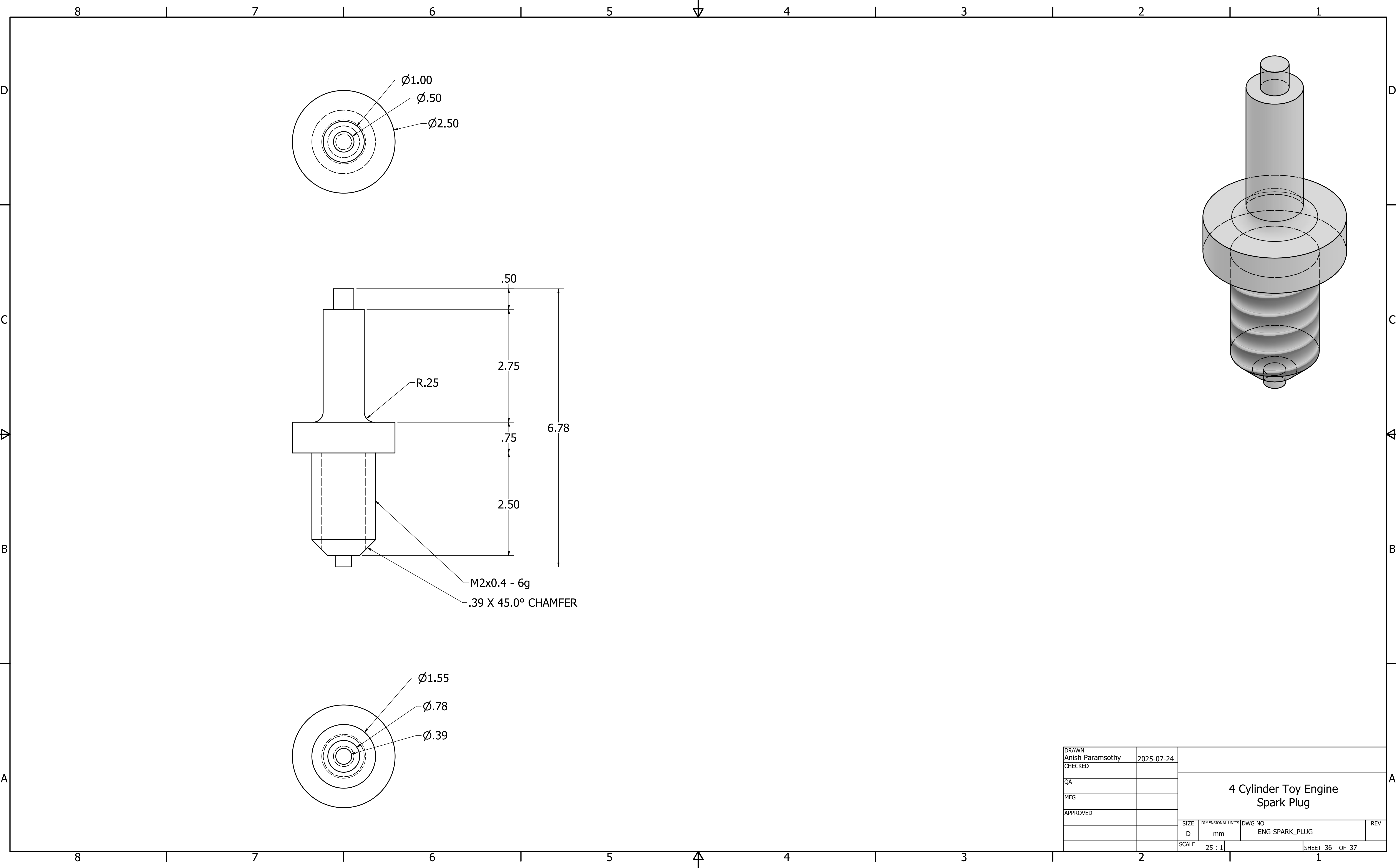
DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Idler Securing Plate		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-IDLER_SECURING_PLATE
			SCALE	30 : 1	SHEET 33 OF 37

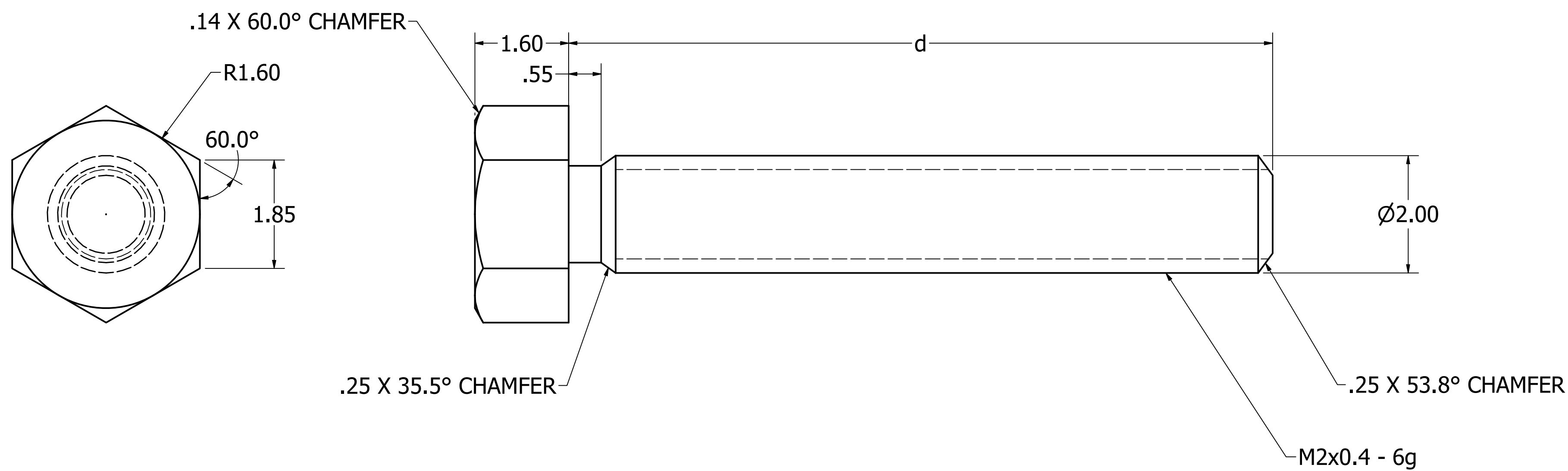
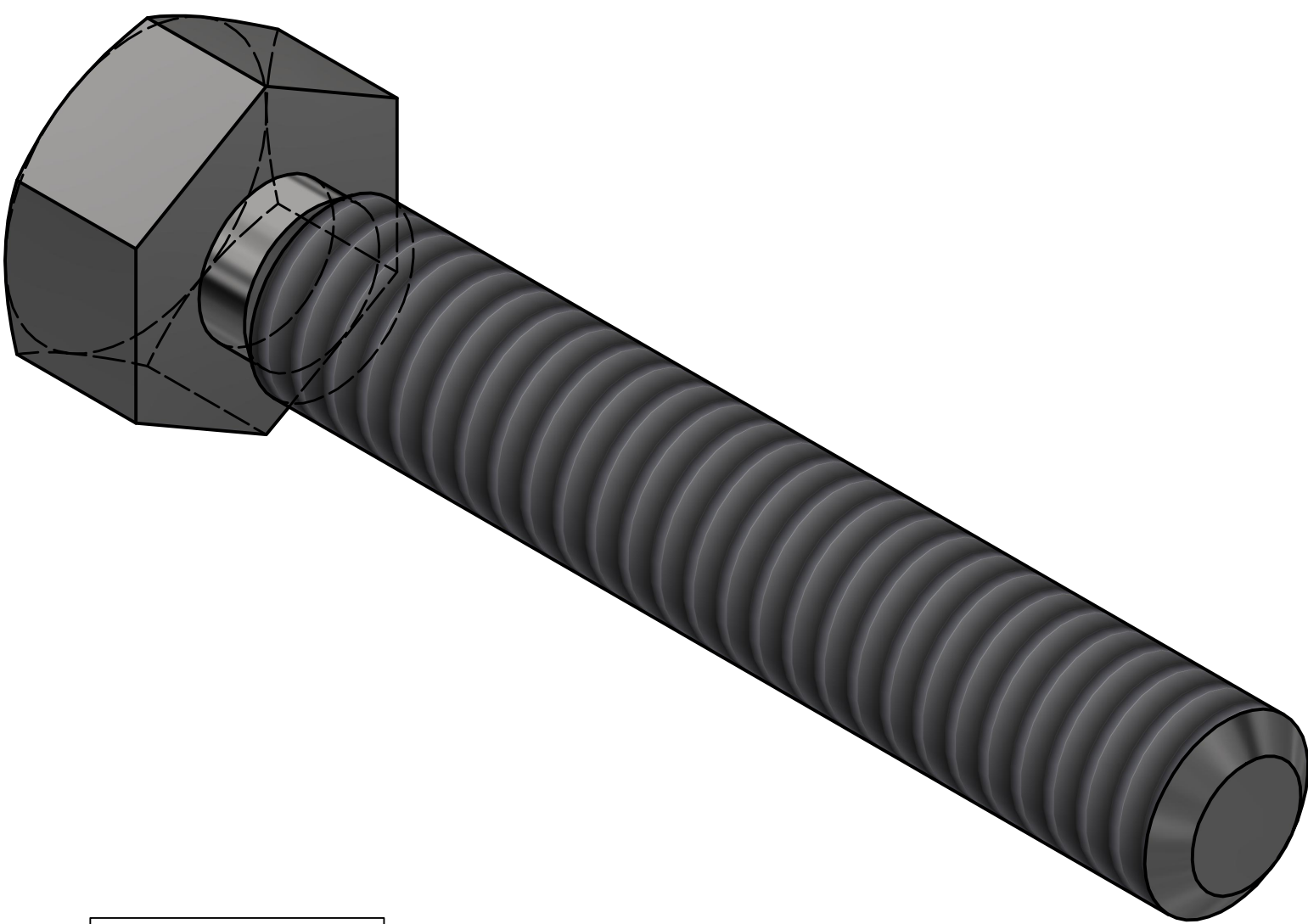


DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Synchronous Belt		
CHECKED					
QA					
MFG					
APPROVED					
			SIZE	DIMENSIONAL UNITS	DWG NO
			D	mm	ENG-SYNCHRONOUS_BELT
			SCALE	4 : 1	SHEET 34 OF 37



DRAWN	Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine Valve Plunger			
CHECKED						
QA			4 Cylinder Toy Engine Valve Plunger			
MFG						
APPROVED			4 Cylinder Toy Engine Valve Plunger			
			SIZE	DIMENSIONAL UNITS	DWG NO	REV
			D	mm	ENG-VALVE_PLUNGER	
			SCALE	20 : 1	SHEET 35 OF 37	





Screw Sizes (M2 x 0.4 x d)	
d	
4	
8	
10	
12	

DRAWN Anish Paramsothy	2025-07-24	4 Cylinder Toy Engine ANSI B 18.6.7 M / IFI 513 - M2 x 0.4 x d THHMSM		
CHECKED				
QA				
MFG				
APPROVED				
		SIZE D	DIMENSIONAL UNITS mm	DWG NO
		SCALE 18 : 1		REV
				SHEET 37 OF 37